NTIA-Report-80-55

Interim Federal Standard 1033 Reference Manual

N.B. Seitz

U.S. DEPARTMENT OF COMMERCE
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December 1980

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PREFACE

This report is the first of a planned two-report series designed to assist users in understanding and applying Interim Federal Standard 1033, "Telecommunications: Digital Communication Performance Parameters." This first volume, the Interim Federal Standard 1033 Reference Manual, outlines potential benefits of the standard, summarizes its objectives and content, and provides a tutorial "essay" on the meaning and importance of each standard parameter. Its sequel, the Interim Federal Standard 1033 Application Manual, will provide guidelines for applying the standard in user requirements analysis, service performance specification, and service selection; and will illustrate the use of these guidelines in a representative system development example.

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INTERIM FEDERAL STANDARD 1033 REFERENCE MANUAL

Neal B. Seitz*

There is a growing need within the Federal government for a user-oriented, system-independent, functional means of specifying data communication performance. A recently published Federal Standard, Interim Federal Standard 1033, defines a set of standard performance parameters designed to meet that need. This report is basically an explanation and elaboration of that standard. The report first outlines the need for the standard, and the potential benefits of its use, from the viewpoint of the end user, the communication supplier, and the communication manager. The report then summarizes the objectives and content of the standard in informal, nontechnical terms. Finally, the report examines the meaning and importance of each standard parameter in a series of tutorial parameter "essays." Typical parameter values are presented, and design implications are discussed.

1. INTRODUCTION

1.1 Background

On March 8, 1979, the Federal Telecommunications Standards Committee (FTSC) voted to approve publication of a new Federal Standard: Interim Federal Standard 1033, "Digital Communication Performance Parameters". The purpose of the new standard is stated in its opening paragraph:

> "to improve Federal government procurement of digital telecommunication systems and services by providing user-oriented, system-independent means of specifying communication performance."

The essence of the FED STD 1033 approach is summed up in the phrase "user-oriented, system-independent". The FED STD 1033 parameters focus on user performance concerns rather than engineering design considerations; they describe end-to-end services rather than particular system facilities; and they apply to all systems, irrespective of transmission medium, network topology, or control protocol. These standard parameters will improve Federal data communication procurement by providing a common framework for functional specification and top-down design; and will promote competition and innovation in the data communications industry by simplifying performance comparison.

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The FTSC approval of FED STD 1033 culminated over 5 years of development, coordination, and review efforts involving more than a dozen Federal agencies. The National Telecommunications and Information Administration's Institute for Telecommunication Sciences (NTIA/ITS) chaired the FTSC's Telecommunication Performance Standards subcommittee, and had overall responsibility for developing the standard (Seitz and McManamon, 1978). The National Communications System (NCS) organized and chaired the FTSC, and played a key role in coordinating the standard within the Federal government and with industry. The National Bureau of Standards contributed importantly to the subcommittee work, and provided early contacts with non-FTSC standards organizations, notably the American National Standards Institute (ANSI). The General Services Administration provided guidance on Federal implementation of the standard and was responsible for its formal publication (GSA, 1979). Other Federal contributors included the National Aeronautics and Space Administration, the Library of Congress, and the Defense Communications Engineering Center.

The FED STD 1033 development effort also benefited substantially from various nongovernment standards activities. Over a dozen industry organizations participated in the FCC's specialized common carrier Quality and Reliability inquiry (FCC, 1975); and their recommendations were summarized in an ITS report which provided a technical foundation for the standard (Seitz and McManamon, 1976). The FTSC maintained close coordination with CCITT and ISO groups addressing performance issues during development of the standard (e.g., CCITT, 1977; ISO, 1978); and the draft standard was aligned with international recommendations on a number of topics. NCS solicited direct public comments on the standard on two separate occasions (NCS, 1977; NCS, 1978); and these comments had a substantial impact on the final document.

Perhaps the most important non-Federal contribution to the standard was that of ANSI's Data Communication Performance Task Group, X3S35. That group has been working in standard performance assessment for over 15 years, and has produced two protocol-based ANSI performance standards (ANSI, 1974; ANSI, 1980). Both standards were valuable precedents for FED STD 1033, and the Task Group's detailed review of the draft Federal Standard substantially improved its clarity. Task Group X3S35 and the FTSC's Telecommunication Performance Standards subcommittee essentially united in 1979, with the objective of adapting and refining FED STD 1033 for proposal as an American National Standard. That work is currently under way.

In approving FED STD 1033, the FTSC also adopted an important qualifying provision - that the standard be designated as an interim Federal Standard. The "interim" designation identifies a Federal Standard as an evolving standard,

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typically undergoing initial application trials; and makes its use by Federal agencies optional. Non-interim federal standards typically are more established, and are mandatory for use by all Federal agencies. Interim federal standards are normally converted to mandatory federal standards after the necessary application trials (and any necessary revisions) are completed.

A number of factors led the FTSC to conclude that the interim step was necessary in the case of FED STD 1033. The first was the relative novelty of the FED STD 1033 approach. The standard places the user/system interfaces well outside the traditional DTE/DCE[1] boundaries; defines the performance parameters in terms of general "reference events" rather than system-specific interface signals; and treats the user and the system as co-responsible entities who jointly determine overall communication performance. These departures from convention clearly suggested the need for a user familiarization period.

A second reason for the interim decision was the substantial impact the standard will have on existing Federal procurement practices. FED STD 1033 requires (and enables) a functional approach to communication procurement, in which the end user needs are specified without presupposing any particular system design. As discussed in Section 2, such an approach is far from realization in many Federal organizations today. The interim period will allow time for necessary procedural changes to occur gradually, in a natural, evolutionary way.

A final reason for the interim decision was the fact that the standard gave at least some reviewers an impression of substantial complexity. The following comment (from an otherwise favorable Department of Defense reviewer) illustrates one such reaction:

> "The standard and supporting documents are judged to be a highly sophisticated technical approach to a difficult problem, which is to treat all manner of digital communication systems under the same standard umbrella. The standard represents the latest in technical thought on the subject and as such is not intended for the uninformed reader. Much effort must be devoted to understanding it."

Although many reviewers did not share this opinion, the fact that a significant minority did reinforced the view that a gradual implementation of the standard would be advisable.

The question of complexity deserves a brief separate discussion here. There is nothing inherently complex about the performance parameters specified in FED STD 1033; they express basic performance concerns which are readily understandable,

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[1] Data Terminal Equipment/Data Circuit-Terminating Equipment.

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and vitally important, to data communications users. The impression of complexity the standard conveys to some is a result of the rather rigorous way the parameters are defined. Such rigor was necessary, in the absence of a completed measurement standard, to ensure the comparability of measured performance parameter values. Two effective steps are now being taken to eliminate this problem:

1. NTIA/ITS is developing a performance measurement standard which will incorporate the mathematical details of the performance parameter definitions into a standard, machine-independent computer program. This standard program will transform observed performance data into parameter values in a totally uniform way, to ensure comparability; and will eliminate the need for mathematical formality in FED STD 1033. The new standard, Federal Standard 1043, will be titled "Digital Communication Performance Measurement Methods."

2. On the strength of the upcoming measurement standard, the joint ANSI/FTSC Task Group is recasting the FED STD 1033 parameter definitions in a more informal, narrative style. The developing ANSI standard, designated X3S35/125, is titled "User-Oriented Data Communication Performance Parameters."

Interim Federal Standard 1033 was officially published by GSA's Federal Supply Service on August 29, 1979. Since that time, NTIA and NCS have received over 600 separate requests for the standard, from government and industry organizations in the U.S. and more than a dozen foreign countries; and a number of initial applications of the standard are under way. It is likely that when the joint ANSI/FTSC Task Group completes its adaptation of FED STD 1033, the U.S. government will adopt the ANSI version as a mandatory Federal Standard, to be used in all Federal data communication procurements. Feedback from initial applications of the interim Federal Standard will be extremely useful to the joint Task Group in shaping its ultimate successor.

1.2 Purpose and Scope of Report

The purpose of this report is to encourage and facilitate initial use of Interim Federal Standard 1033 by providing an informal, non-technical presentation of its objectives and content. Earlier reports on 1033 were directed to standards developers, and addressed technical issues associated with parameter definition and measurement. This report is directed to standards users, and presents the standard from a more practical, user-oriented perspective.

The report is divided into three major sections. Section 2 outlines the need for the standard and the potential benefits of its use. Section 3 summarizes the objectives and content of the standard. Section 4 provides a tutorial "essay"

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on the meaning and importance of each standard parameter. An annotated bibliography of technical reports and papers dealing with performance assessment issues is provided in Section 5. Guidelines for applying the standard in actual communication procurements will be provided in a planned sequel to this report, the Interim Federal Standard 1033 Application Manual.

The explanatory, user-oriented nature of this report necessarily imposes certain limitations. The report takes the FED STD 1033 parameters as a given starting point, and provides little discussion of how or why they were selected. The parameter selection process is described thoroughly in Seitz and McManamon (1978). The report defines the standard parameters in an informal, narrative style, with mathematical details and "fine points" intentionally omitted. More rigorous parameter definitions are provided in the standard itself. Finally, the report does not address the complex subject of performance measurement, or related issues such as sampling strategy. These topics will be addressed in the measurement standard (FED STD 1043) and in later reports. A technical basis for this work has been established in a series of ITS reports by Crow (1974, 1978, 1979) and Crow and Miles (1977).

2. THE BENEFITS

2.1 Introduction

Why use Federal Standard 1033? The question is a serious one, particularly since adherence to the standard will require a significant change in procedures and thinking for many potential users. This section answers that question in terms of tangible benefits that will accrue to end users, suppliers, and communication managers who use FED STD 1033 as a functional framework for future communication procurements. The section is divided into three major subsections. The first outlines existing Federal data communication procurement practices; the second identifies current sources of inefficiency in these practices; and the third describes the benefits of the functional procurement methods proposed herein. Current inefficiencies and potential benefits are discussed from three points of view: end user, supplier, and communication manager. Although the discussion focuses on Federal procurement, similar problems and opportunities exist in many non-Federal organizations as well.

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2.2 Existing Procurement Practices

A useful description of existing Federal data communication procurement practices requires that we distinguish three general categories of participants: end users, communication suppliers, and communication managers. An end user is an individual or entity that produces or "consumes" information transmitted over a telecommunication system. Typical end users of data communication service are a human terminal operator and a remote computer application program in a teleprocessing network. We employ the term end user (or "user") rather broadly here, to denote either groups of users or individuals.

The term communication supplier here denotes any nongovernment organization that provides telecommunication services or equipment to the Federal government. Traditionally, the major suppliers of telecommunication services have been the common carriers. The equipment suppliers include suppliers of interconnect equipment (e.g., modems, data terminals, PBX's); and suppliers of primary equipment (e.g., microwave and satellite terminals, antennas). Inasmuch as all communication equipment exists to provide communication services, it can be said that all communication suppliers ultimately contribute to the provision of services.

The third category of participant in Federal communication procurement is the communication manager. The communication manager essentially serves as a broker, or middleman, between a user who requires communication services and a supplier or group of suppliers who provide them. There are at least three levels of communication managers in the Federal government: (1) organizational managers, responsible for a particular installation or activity (e.g., the Department of Commerce Laboratories in Boulder); (2) departmental managers, responsible for an overall Federal agency (e.g., the Department of Commerce); and (3) administrative managers, responsible for coordinating multi-agency procurements. The latter function is performed for most nonmilitary agencies by GSA's Automated Data and Telecommunications Service (ADTS), and for the Department of Defense by the Defense Communications Agency (DCA). At the organizational level, the communication management function typically is performed by a single individual; at the administrative level, the function is performed by a large Federal organization with a multi-million dollar budget.[2]

These characterizations provide a basis for describing the key steps in a typical Federal data communication procurement. For the sake of concreteness, we

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[2] As an example, GSA/ADTS employs 2,500 people and buys $500 million worth of data communications equipment annually.

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presume a situation in which data communication service is needed to interconnect the users of a number of geographically remote time-sharing computers owned by a single, nonmilitary Federal organization. The procurement would be conducted, according to GSA's Federal Property Management Regulations (FPMR's), as follows (GSA, 1978):[3]

1. The need for a data communication service is perceived by a group of users (e.g., programmers, administrators, ADP managers) at the various computing sites. As an example, frequent conferences and close working relationships between individuals at different work sites might require the exchange of large amounts of digital information, in real-time, between sites. Or, two or more remote computers might serve as backups to each other, to maintain data processing service continuity during periods of equipment outage or unusually heavy usage.

2. Representatives of the various user communities organize a "data communications study group" which conducts a "data communications study." Quoting from Subchapter F, Part 101-36 of the FPMR's,

   > "The data communications study includes a detailed analysis of the proposed data processing system and the environment within which it will operate and a determination as to the feasibility and economy of data communications under the circumstances. Also, such a study indicates the additional equipment and the type and number of communications lines which are estimated to be required; the impact on the format of data and data banks, codes to be used and programming required; and, most significantly, the important elements of cost."

3. The study group produces, as its output, "a written report detailing the data communications system which most economically and effectively satisfies the requirements of the proposed data processing system."

4. The report is transmitted to a higher authority (with responsibility for both the individual computers and the proposed communications extension) for analysis and validation. The FPMR's state that "such analysis should consider the extent to which the proposed data communications system will satisfy the requirements of the proposed data processing system and should include a cost/benefit study to determine whether to include or exclude data communications from the proposed data processing system."

5. Upon completion of this review, the organizational communication manager prepares a series of telecommunication service request forms (GSA 2936-B) detailing the required circuits and facilities; and transmits these forms, through the appropriate departmental communication manager, to GSA.

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[3]The discussion disregards the agency process of funding acquisition.

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6. GSA/ADTS communication management personnel review the service request forms and approve or disapprove them within 20 workdays. In this connection, the FPMR's state that "if no action is taken by GSA within the 20 workdays after receipt of a request from an agency ... the agency concerned may proceed as if, in fact, approval had been granted."

7. Upon receipt of GSA approval, the organizational communication manager submits an order for the necessary communication equipment and services to the appropriate government procurement officials; and the normal process of supplier bidding, bid evaluation, contract award, delivery, and installation proceeds from there.

The DoD procurement process differs from that described above in various details, but the basic plan is similar. Requirements for new services are perceived and initially reviewed within the using commands. Each military department has one or more validation offices, equivalent to the departmental communication managers, that are responsible for assessing new requirements. Validated requirements are submitted to the Defense Commercial Communications Office of DCA, which accomplishes the actual procurement of services from nongovernment suppliers (GAO, 1977).

2.3 Sources of Inefficiency

Any system specification process involves two fundamental steps:

1. Specifying what the system must do, in terms of a set of required functions and associated performance levels;

2. Specifying how the system will achieve these objectives, in terms of specific components, interconnections, and operations.

The first step is alternatively called a "user requirements analysis" or "system requirements analysis." Properly conducted, it is a careful examination of the user function the system must support; it determines the quantitative impact of system performance on user effectiveness, and thereby defines the objectives of system design. As an example, communication delay would have vastly different impacts on the user functions of inventory accounting and missile fire control; and the user requirements in the two cases would differ correspondingly. The output of the requirements analysis step is called a functional specification to emphasize the fact that it defines "what", but not "how".[4]

The second step in the system specification process is the detailed system design. In this step, the analyst postulates various ways a system could be constructed to meet the user requirements; and evaluates each relative to the

[4]Such a specification also identifies relevant cost and operational constraints.

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defined constraints. As an example, a given end-to-end delay requirement might be satisfied by a dedicated communication link with a relatively low information transfer rate; a polled message switching network with a higher rate; or, conceivably, by a totally nontelecommunications solution such as express mail. The output of the system design step is called a design specification (or fabrication specification) to emphasize its focus on "how." In sum, the functional specification defines the service; the design specification defines the system that provides that service.

A major deficiency in current Federal data communication procurement practices is that they do not clearly distinguish between these two basic specification steps. The "data communications study" described in the FPMR's includes both the user requirements analysis and the system design; its output, the study report, documents the results of both. No intermediate functional specification is required or, in fact, even suggested.

If anything, the current FPMR's encourage agencies to mix the specification of user requirements with system design. As an example, one of eight "factors to be examined" in a data communication study is described as follows: "Accuracy required and the necessity for employment of error detection and correction techniques." Accuracy is a user performance requirement; error detection and correction techniques are particular methods of achieving a stated accuracy requirement. The FPMR's list the following factors to be identified in the data communications study report:

1. Type of service.
2. Line requirements.
3. Hardware and software requirements.
4. Error detection/correction techniques required.
5. Time- or event-dependent statement of additions to the communications capability required for the handling of expected increases in workload or in demands on the system.
6. Space requirements.
7. Detailed statement of estimated one-time and recurring costs.
8. Cryptographic security requirements.
9. Noncryptographic security requirements.

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10. Rationale why data communication via electronic means is required as opposed to mail or courier service.

11. Access security.

12. Other facilities in the area which can accommodate the requirement within their existing or enhanced capabilities.

13. Review of existing communications capability.

Factors 1-4, 7, 12, and 13 are outputs of a system design, i.e., a study of "how". The other seven factors are outputs of a user requirements analysis, i.e., a study of "what".

Combining user requirements analysis and system design in a single "data communications study" often has the effect of forcing the user to assume the primary responsibility for system design. Although the study group can, in principle, include communication managers and other consultants as well as users, it is the user who needs the service; and necessity motivates participation. At the same time, current GSA regulations do not allow the study group to stop at the functional specification stage. As noted above, the FPMR's stipulate that the study group must produce a detailed design specification, and obtain GSA/ADTS approval of it, prior to procurement. Figure 2-1 identifies the categories of information such a group would submit to GSA at Step 5 of the procurement process described above; in essence, what is required is a list of specific services and equipment needed to implement a completed system design.

Forcing the user into a design role has a number of major disadvantages. These can be described from the point of view of the end user, the supplier, or the communication manager. The typical end user views data communication as transport service to be used, like the mail, in moving information from one place to another. He has little interest in how information transport is physically accomplished; his concerns are with its ultimate speed, accuracy, and reliability, and with the ultimate cost of the service.

Most users would be very willing to stop at the functional specification stage if they could be sure their requirements would be met in a reasonable manner. Communication system design to them is a complex, time-consuming, treacherous process which diverts valuable time and resources from their real mission. They feel, justifiably, that they should not have to understand the design of a system in order to use it.

Suppliers approach the data communications market place as sellers rather than as buyers, but their dissatisfaction with user design is no less strong. To a

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Data, Facsimile, and Record Telecommunications Services Serving Computers

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SECTION I – GENERAL INFORMATION

1. Transaction: A Add; C Change
2. Submission Type: F FPMR Approval Request; N Notification
3. Organization Code
4. System ID
5. Operation Date: YR MO
6. Disconnect Date: YR MO
7. Submission Date: YR MO DY
8. Will the requested facilities be used for the storage, retrieval, processing, or transfer of personal information? If so, please attach appropriate certification as prescribed in Federal privacy regulations. Y Yes; N No

SECTION II – COMPUTER LOCATION INFORMATION

9. Location ID
10. Operating Activity *(Optional)*
11. Street Address
12. City
13. State
14. ZIP Code

SECTION III – CIRCUIT AND MODEM INFORMATION

CIRCUIT INFORMATION								MODEM INFORMATION				
Reference Line No. (a)	Quantity (b)	Common Carrier Code or Name (c)	Restoration Priority (d)	Type of Circuit (e)	Full or Half Duplex (F or H) (f)	Conditioning (g)	Local or Intercity (L or I) (h)	Hard Wired or Acoustic Coupled (H or A) (i)	Manufacturer Code or Name (j)	Model No. (k)	Purchased or Leased (P or L) (l)	Cost Per Modem in Dollars *(Optional)* (m)	Transmission Speed in Bits Per Second *(BPS)* (n)

15. Remarks *(If necessary, continue on reverse)*

16A. REQUESTOR'S SIGNATURE | 16B. REQUESTOR'S NAME AND TITLE *(Type or print)* | 16C. TELEPHONE NO.

16D. AGENCY AND OFFICE | 17A. CONTACT'S NAME | 17B. TELEPHONE NO.

GENERAL SERVICES ADMINISTRATION — GSA FORM 2936-B (4-76)

Figure 2-1. Service specification form.

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supplier, the existence of a preconceived user design in a procurement specification means two things: (1) his opportunity to choose the most efficient, economical method of meeting the user need has been usurped; and (2) he may have been precluded from seeking the user's business altogether if the products or services of a competitor have been specified in preference to his own. User-developed design specifications are frequently incomplete, ambiguous, and misleading; and such specifications inevitably increase supplier uncertainty, inefficiency, and financial risk. Poorly prepared specifications can also directly reduce supplier revenue, both by delaying the installation of new services and by hastening their abandonment. In short, user design efforts are often most unwelcome to the data communication supplier.

As noted earlier in this section, the Federal communication manager essentially operates as a broker between a user who requires communication services and one or more suppliers who provide them. Federal communication managers are charged with two general objectives: (1) to meet the communication needs of their user clients at the lowest possible cost, and (2) to implement Federal policy guidelines in such areas as reliance on the private sector (OMB, 1979); efficient spectrum utilization (NTIA, 1980); and Federal Telecommunications System interoperability (National Security Council, 1979).

Both communication management responsibilities are frustrated by the current concentration of system design responsibility in Federal user organizations. Since the typical user has relatively little communication expertise, his "designs" tend to be inefficient, costly, brute-force approaches - e.g., dedicated lines. Faced with the complex task of designing a communication system, users tend to skip the requirements phase altogether, with the result that the procured system may have little relationship to actual needs. Even in cases where users are capable of efficient design, they are not in a position to assess the feasibility of "common user" solutions, since they are not normally aware of the requirements of other Federal users. Once a user organization has committed itself to a particular system design, that design becomes the basis of discussion with the communication manager, with the result that little consideration is given to other design alternatives or applicable Federal policy guidelines. In extreme cases, the communication manager is reduced to a "paper pusher" and "front man" with no real authority or responsibility for anything.

Users placed in a design role often seek direct assistance from communication suppliers. While most suppliers are undoubtedly conscientious, they have little

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incentive to suggest alternatives provided by their competitors; and they may be unaware of pertinent Federal policy guidelines. Suppliers placed in a communication management role also have a strong tendency to overdesign: first, because they are uncertain about the real user need; and second, because they receive more revenue from more elaborate services. Like the users, suppliers often are not in a position to aggregate requirements among different Federal organizations. Premature user consultations with suppliers often result in the user's requirements being very simply defined: "whatever the last salesman out the door said he needed."

The following excerpts from a recent General Accounting Office report substantiate these claims with respect to defense communications (GAO, 1977):

- "Users, rather than a centralized authority having cognizance of Defense-wide needs, decide upon the method of satisfying their individual requirements."

- Requests for new services often lack sufficient explanatory information to permit adequate consideration of alternative means of satisfying the requirements. Validation officers we talked with indicated that users' requests for dedicated services are seldom questioned if the user can provide the funds."

- "Defense Commercial Communications Office officials contend that they are not technically or quantitatively staffed to prepare designs or provide alternative solutions to service from those proposed by the carrier. Air Force Communications Service officials informed us that their review of the carrier's proposal was limited to determining its technical adequacy."

- "Based on our review of about 500 leased dedicated circuits costing over $5.6 million annually, we found about 450 circuits, costing nearly $4.9 million annually, which are candidates for either elimination, reconfiguration for more economical service, or integration into switched common-user networks."

This situation raises some fundamental questions. Data communication procurements are going on at a rate approaching $2 billion annually in the Federal government. If these procurements are, in fact, poorly conceived in a large number of cases, how much potential improvement in Federal user productivity is being sacrificed? How much is industry innovation being inhibited? How much Federal tax money is being wasted on inappropriate system configurations?

These questions are difficult to answer in quantitative terms, but the impacts clearly are substantial in all areas. In commenting on ITS projections of potential savings in the latter area, a prestigious National Research Council study panel made the following observation (NRC, 1977):

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> "The impact of a good data network selection method for the General Services Administration (GSA) to meet the users' needs at the least cost is understated at best. The impact would not be just the quoted 5% of 6% of the government data bill ... A larger goal to shoot for is avoiding the wasteful procurement of a system significantly more costly than necessary to fill the requirement. If we conservatively assume 20% could be saved by an efficient method to select the right service or system, the impact could be $400 million per year ...."

Federal data communication expenditures are currently growing at a rate in excess of 20% per year, and are expected to continue to do so at least through the mid-1980's.

2.4 FED STD 1033 Benefits

Why is the distinction between user requirements analysis and system design so unclear in current Federal procurement regulations? Why has it been so difficult to realize the benefits of a functional approach to data communication procurement? A key problem has been the absence of a suitable framework for functional performance specification - in essence, a "common language" for relating the performance needs of end users with the capabilities of supplier systems and services.

Figure 2-2 compares the existing and functional procurement approaches and illustrates the crucial role of the functional specification in achieving the latter. The existing approach, Figure 2-2a, mixes user requirements analysis and system design in a single, undifferentiated "data communication study." This approach encourages a mixing of "what" with "how", and results in poor delineation of the user requirements and an inefficient, costly system design. The functional approach, Figure 2-2b, clearly distinguishes the user requirements analysis and the system design as separate studies; and produces a clear, complete description of the user requirements and a more cost effective system design. The key element that makes this separation possible is the functional specification: a single statement of both the user requirements and the minimum system objectives in terms of system-independent, functional performance descriptors. Federal Standard 1033 provides such a set of functional performance descriptors.

Just how will FED STD 1033 benefit Federal data communication procurement? Again, the question can be addressed from the point of view of the end user, the

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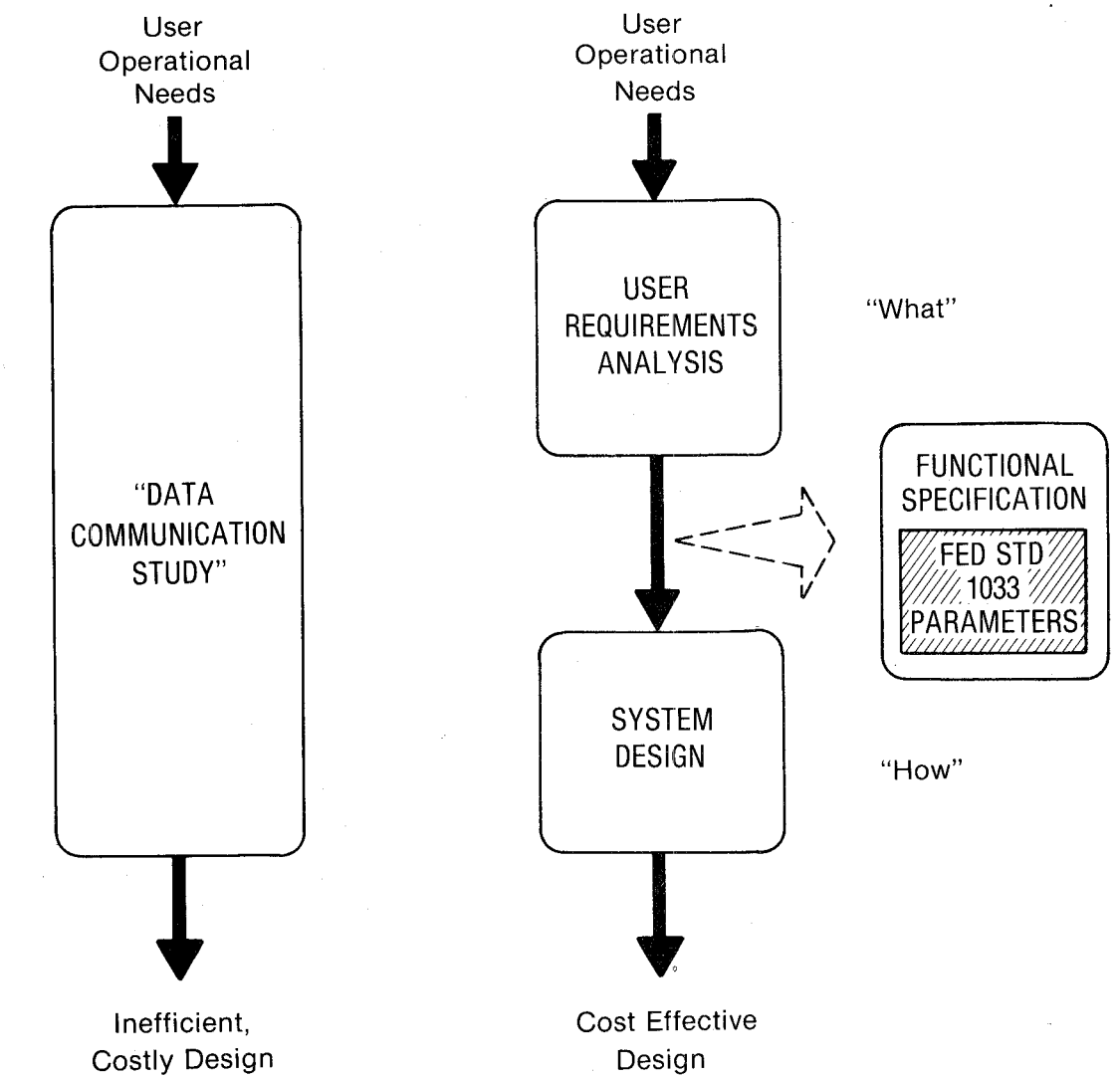


Figure 2-2. Existing vs functional procurement.

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supplier, or the communication manager.[5] The standard will benefit Federal end users in two obvious ways:

1. By relieving them of a burdensome responsibility for communication system design. Users will be enabled and encouraged to regard data communications as a transport service - their natural inclination in the first place.

2. By allowing them to define their data communication needs more precisely. The 1033 parameters will enable users to pinpoint the specific impacts of communication performance on their own operations, thereby minimizing procurement uncertainty and the risk of costly mistakes.

An equally beneficial, but less obvious, user application of FED STD 1033 will be in assessing potential uses of data communications in improving government productivity. Federal employees frequently encounter situations where it seems clear that a necessary government function is "information limited" - i.e., the function could be performed much more efficiently, with a substantial cost savings to Federal taxpayers, if accurate, timely input information were available. Nevertheless, it can be very difficult to quantify the potential savings in monetary terms; and Federal budget planners quite properly expect such evidence if a major commitment of resources is required. Federal Standard 1033 provides an ideal framework for assessing such situations because it focuses analysis on the user need without presupposing a particular design solution - telecommunications or otherwise. The effect of information delay, inaccuracy, or unavailability on an information-dependent function can be determined without specifying a transport method for that information; and the FED STD 1033 parameters enable the analyst to do just that.

Federal Standard 1033 will benefit data communication suppliers in three tangible ways:

1. By enlarging their participation in the design of Federal data communication systems. Suppliers will receive both the direct stimulus of new "turn-key" business, and the indirect stimulus of enhanced design responsibility.

2. By simplifying the process of describing their system or service offerings. Suppliers will be able to develop a single basic performance specification applicable to all potential Federal uses.

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[5]The discussion assumes supplier design responsibility. Similar benefits accrue when Federal communication managers are responsible for design as long as a system-independent functional specification is developed.

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3. By maximizing their opportunity to compete for Federal business. Expanded use of functional specifications in Federal procurement will prevent arbitrary exclusion of qualified suppliers.[6]

Use of the standard will also benefit data communication suppliers in an indirect way, by rewarding successful innovation and the marketing of cost effective products and services.

Federal Standard 1033 will assist Federal communication managers in discharging both their user service and their policy implementation responsibilities. To meet a user's data communication needs, a communication manager clearly must know what those needs are - and few managers enjoy that knowledge under the present conditions. Implementing Federal policy guidelines requires a certain authority over Federal procurement decisions - and again, few Federal communication managers actually have that authority today. Successful implementation of the standard will provide specific benefits in at least three phases of communication management:

1. Requirements Specification. The standard will improve the quality of user requirements specifications, and will facilitate their development through the cataloging of similar prior applications.

2. Service Acquisition. The standard will simplify the matching of end user requirements with offered systems and services. The NRC estimate cited earlier provides an indication of the total potential improvement here.

3. Service Assessment. The standard will provide a basis for agreement between users and suppliers on the quality and reliability of delivered services, and will facilitate direct comparison of service alternatives.

All of these benefits are a result of the standard's "common denominator" property.

An "ideal" situation with respect to Federal procurement of data communication systems and services might be briefly summarized as follows:

1. Federal User Organizations would understand and accept the need to describe their communication requirements in a system-independent manner. Users would specify performance requirements in functional terms, without reference to particular communication facilities or services. Individual parameter values would be determined on the basis of their impact on the user process being served; as an example, the Bit Error Probability requirement for a digital air traffic control system would be determined by considering the impact of bit errors on air traffic control effectiveness.

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[6]This is true of equipment suppliers as well as service suppliers, since subsystems can also be specified in functional terms.

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2. Industry Suppliers of communication systems and services would be willing to specify their performance in uniform functional terms, and would be appropriately compensated for their effort in doing so. Available facilities and services would be catalogued, along with major Federal applications, in a central communications catalog which would facilitate the design and procurement process.

3. Federal Communication Managers would assist end users in defining realistic communication requirements; and would have the authority and the resources to select the best available means of meeting these needs. Where appropriate, they would aggregate independent user requirements to be met by a single common user system. They would, in sum, effectively manage Federal communication procurement. A major improvement in the cost effectiveness of Federal data communication systems would result.

Federal Standard 1033 will contribute substantially to the realization of this ideal by providing a system-independent, functional framework for future communication procurements.

3. FED STD 1033 EXPLAINED

3.1 Introduction

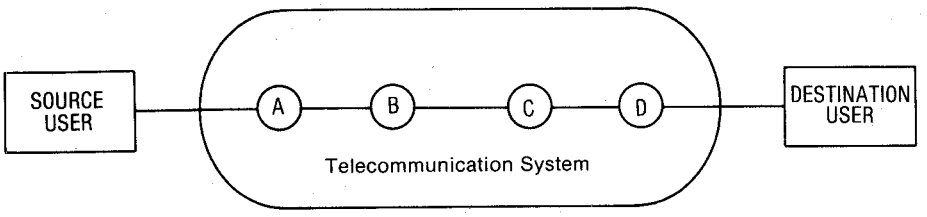
This section summarizes the objectives and content of interim Federal Standard 1033. The section is divided into two major subsections. The first describes three key technical problems which influenced development of the standard. The second summarizes the overall FED STD 1033 approach. Refer to the standard and its supporting reports for further detail on each topic.

3.2 Key Technical Problems

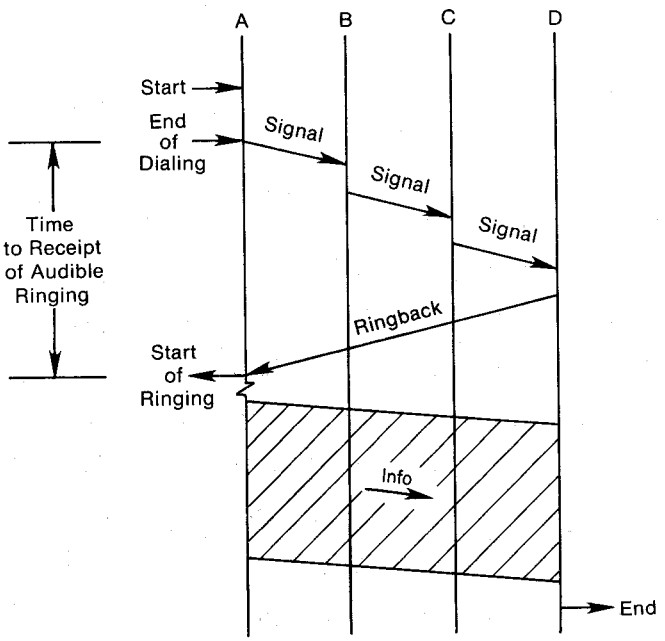
Development of FED STD 1033 required the solution of three key technical problems. The first of these was the problem of system dependence. A survey of candidate parameters revealed that the great majority were defined such that they could only be applied to systems with particular topology or protocol features. This is undesirable because it prevents use of the parameters in comparing systems that provide the same ultimate service by means of different detailed designs.

A good insight into the problem of system dependence can be obtained by considering the differences between traditional circuit-switched and message-switched transactions, as shown in Figure 3-1. (Note that time proceeds down the page in this and similar diagrams). In a circuit-switched transaction, service is provided by setting up an end-to-end path, or circuit, from source to destination

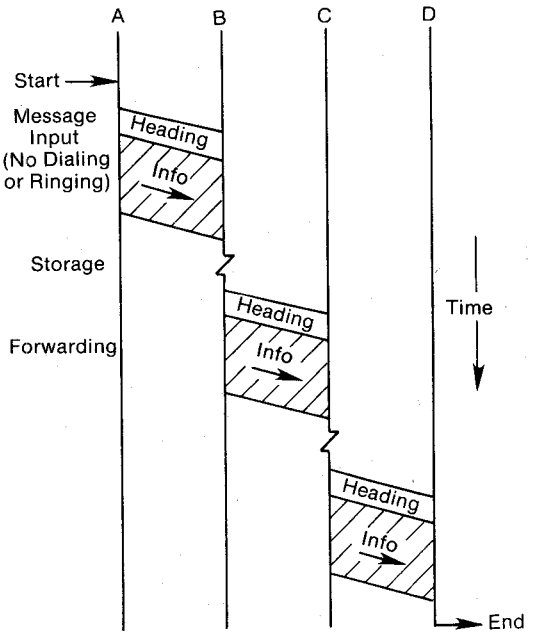
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a. Network Topology.



b. Circuit-Switched Transaction.



c. Message-Switched Transaction.

Figure 3-1. System dependence.

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prior to the start of user information transfer. The individual links which comprise the end-to-end circuit are all allocated to that particular user pair for the duration of the transaction, independent of usage; and all links are used concurrently during transfer. A familiar example is a normal voice telephone call.

In a message-switched transaction, no end-to-end circuit is set up prior to the start of user information transfer. Instead, the user message is forwarded through the network link by link; and the entire message is stored for some period of time at each intermediate node. Individual links are allocated to a particular user pair only during actual forwarding of their message. At all other times, the link may support other users. DoD's AUTODIN I is a classic example of a message-switching system. Another example is GSA's Automatic Record System.[7]

One parameter which is commonly used in expressing the performance of circuit-switched systems is the Time to Receipt of Audible Ringing - the elapsed time from the end of dialing to the start of ringing (Fig. 3-1). No counterpart to this parameter is possible in the case of message-switched systems. The function of switching is performed by interpretation of the message heading at each node, rather than by pre-transmission signaling; and the concepts of dialing and ringing are thus irrelevant. A user wishing to compare the performance of circuit-switched and message-switched services cannot do so in terms of Time to Receipt of Audible Ringing; other, system-independent descriptors of performance are required. The absence of such system-independent performance descriptors has been a major difficulty with performance comparison heretofore.

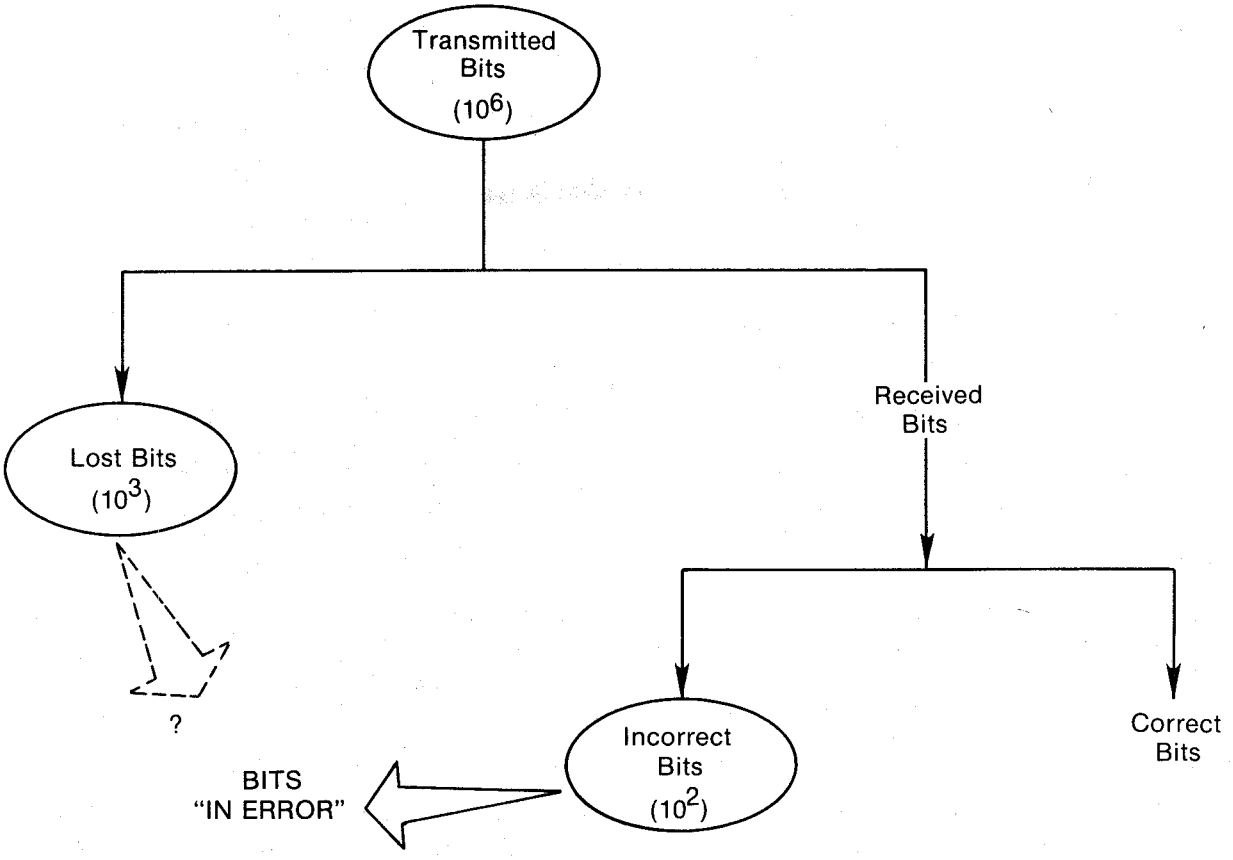
The second key problem encountered in developing FED STD 1033 was the problem of detailed parameter definition. In most cases, traditional narrative definitions of performance parameters are not precise enough to ensure their uniform application to comparable service offerings. The result, of course, is a potential for inefficiency and error in the process of matching service offerings with end user needs.

As an example of the parameter definition problem, consider the familiar accuracy parameter Bit Error Probability (Figure 3-2). A typical narrative defines this parameter as "bits in error per bits transmitted", but makes no mention of whether (or how) bits lost in transmission should be counted. Two obvious choices, both consistent with the narrative definition, would be (1) to count lost bits

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[7] Packet-switching is similar to message-switching, except that the messages are divided into smaller units, called packets, which are forwarded through the network separately.

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$$\left\{\begin{array}{c}\text{Bit Error}\\ \text{Probability}\end{array}\right\} = \left\{\begin{array}{c}\text{Bits "In Error"}\\ \text{per Bits Transmitted}\end{array}\right\}$$

Choice 1 — Lost Bits Counted:

$$\left\{\begin{array}{c}\text{Bit Error}\\ \text{Probability}\end{array}\right\} = \frac{10^2 + 10^3}{10^6} = 1.1 \times 10^{-3}$$

(Disparity in Values)

Choice 2 — Lost Bits Not Counted:

$$\left\{\begin{array}{c}\text{Bit Error}\\ \text{Probability}\end{array}\right\} = \frac{10^2}{10^6} = 10^{-4}$$

Figure 3-2. Detailed parameter definition.

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with incorrect bits in calculating Bit Error Probability; and (2) to consider only received incorrect bits in calculating Bit Error Probability.

This ambiguity can have a substantial effect on measured parameter values. Assume that of a million (10^6) bits transmitted during a test, a thousand (10^3) are lost and a hundred (10^2) are inverted in transmission. The measured Bit Error Probability values under the two choices are 1.1×10^{-3} and 10^{-4} -- an order of magnitude error in interpreting the meaning of a narrative parameter definition.[8] Other examples could be cited, but the problem is clear: defining words in terms of other words is an endless process which inevitably leaves room for misinterpretation.

The third key problem encountered in developing FED STD 1033 was the problem of user dependence. In most cases, the communication process involves a sequence of interactions between the users and the system; and overall communication performance depends, then, on user performance as well as system performance. There is an obvious problem in employing user dependent parameters in specifying required system performance: the carrier or other supplier normally has no control over user performance, and hence cannot ensure that user dependent parameter values will be met. Nevertheless, many of the parameters which best describe communication performance from the end user point of view are user dependent (e.g., "throughput").

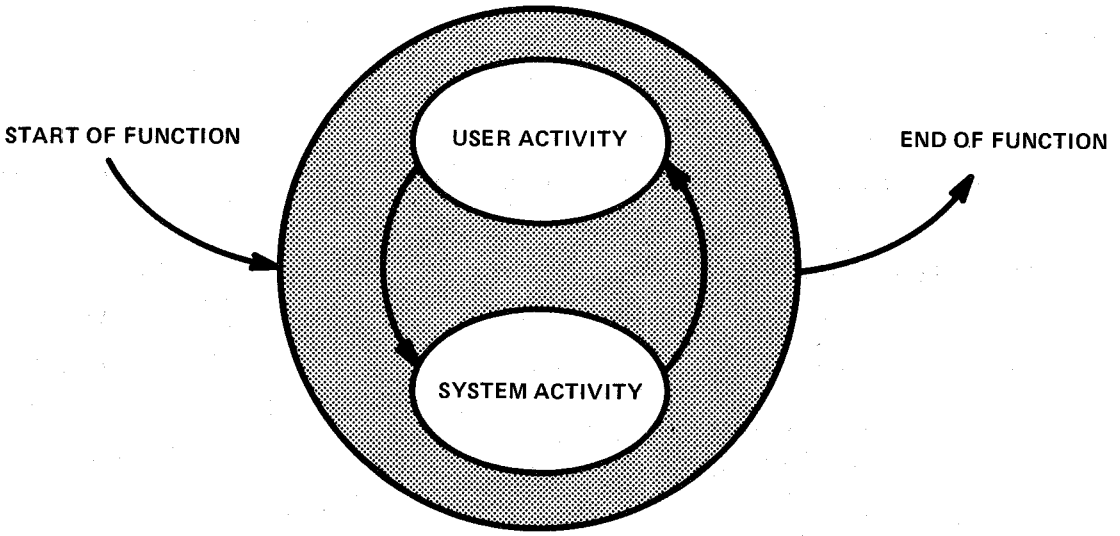
As a simple illustration of the user dependence problem, consider the position of a user who wishes to place a voice call over the public switched network (Figure 3-3). As he initiates the call, his major concern is with how soon conversation can begin, i.e., the total delay between his off-hook action and the called party's answer. The performance parameter Access Time describes exactly this delay; but its values depend not only on the system's speed in signaling and switching, but on the user's speed in dialing and answering.

The telephone companies have traditionally avoided this problem by focusing on parameters which describe unilateral system performance, e.g., Dial Tone Delay and Time to Receipt of Audible Ringing. Unfortunately, such parameters have two major disadvantages from the user point of view: (1) they are system dependent, as noted above; and (2) they do not reflect differences in the "functional burden" placed on the user by otherwise equivalent services. As an example, neither parameter would reflect, in terms of better performance values, the significant

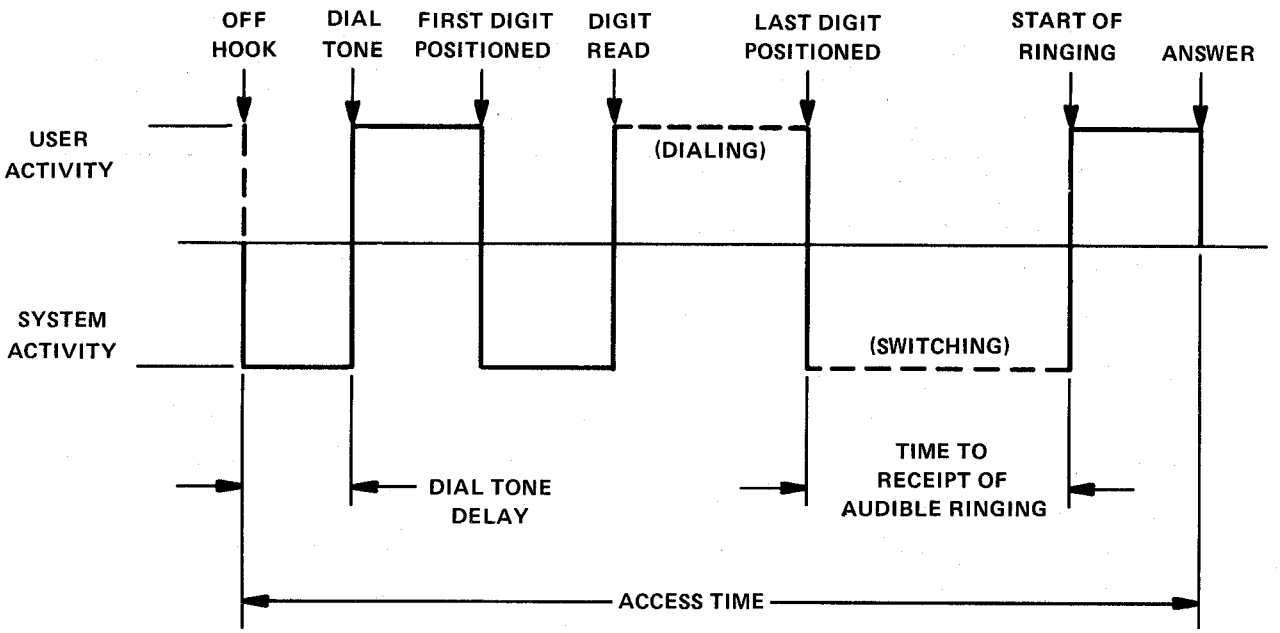
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[8] Data loss does occur in many systems, as a result (e.g.) of signal fading and retransmission protocol failures. See Section 4.3.4 for further discussion.

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a. Alternating Activities within a User Dependent Function



b. Voice Telephone Access Illustration

Figure 3-3. User dependence.

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advantage of abbreviated dialing over conventional rotary dialing. A few recent telephone company studies have recognized this limitation, and stress the need to describe the influence of user delay on overall end-to-end performance (e.g., see Duffy and Mercer, 1978). Nevertheless, a precise quantitative framework for expressing this influence has not been proposed heretofore.

3.3 FED STD 1033 Approach

Figure 3-4 summarizes the overall approach used in developing performance parameters for interim Federal Standard (Seitz and Bodson, 1980). The parameter development process consisted of four major steps:

1. Model Development. Existing and proposed data communication services were surveyed and certain universal performance characteristics shared by all were identified. These characteristics were consolidated in a simple, user-oriented model which provided a system-independent basis for the performance parameter definitions.

2. Function Definition. Five primary communication functions were selected and defined in terms of model reference events. These functions (access, bit transfer, block transfer, message transfer, and disengagement) provided a specific focus for the performance description effort.

3. Outcome Definition. Each primary function was analyzed to determine the possible outcomes an individual "trial performance" might encounter. Possible outcomes were grouped into three general outcome categories: successful performance, incorrect performance, and nonperformance. These categories correspond to the three general performance concerns (or "criteria") most frequently expressed by end users: efficiency (or "speed"), accuracy, and reliability.

4. Parameter Selection. Each primary function was considered relative to each performance outcome in matrix fashion; and one or more specific parameters were selected to represent performance relative to each function/outcome pair. Parameters were selected on the basis of expressed user interest, and consisted of probabilities, waiting times, time rates, and rate efficiencies. The matrix approach ensured that no significant aspect of communication performance would be overlooked in the parameter selection process.

The following paragraphs describe the results of these steps in more detail.

3.3.1 Model Development

In order to describe communication performance as seen by the end user, it is necessary to develop a "user's-eye view" of the communication process itself. What is the nature of the interface between an end user and a telecommunication system, and how is information transferred across such interfaces? How can the process of telecommunication be described in a way that is meaningful and familiar

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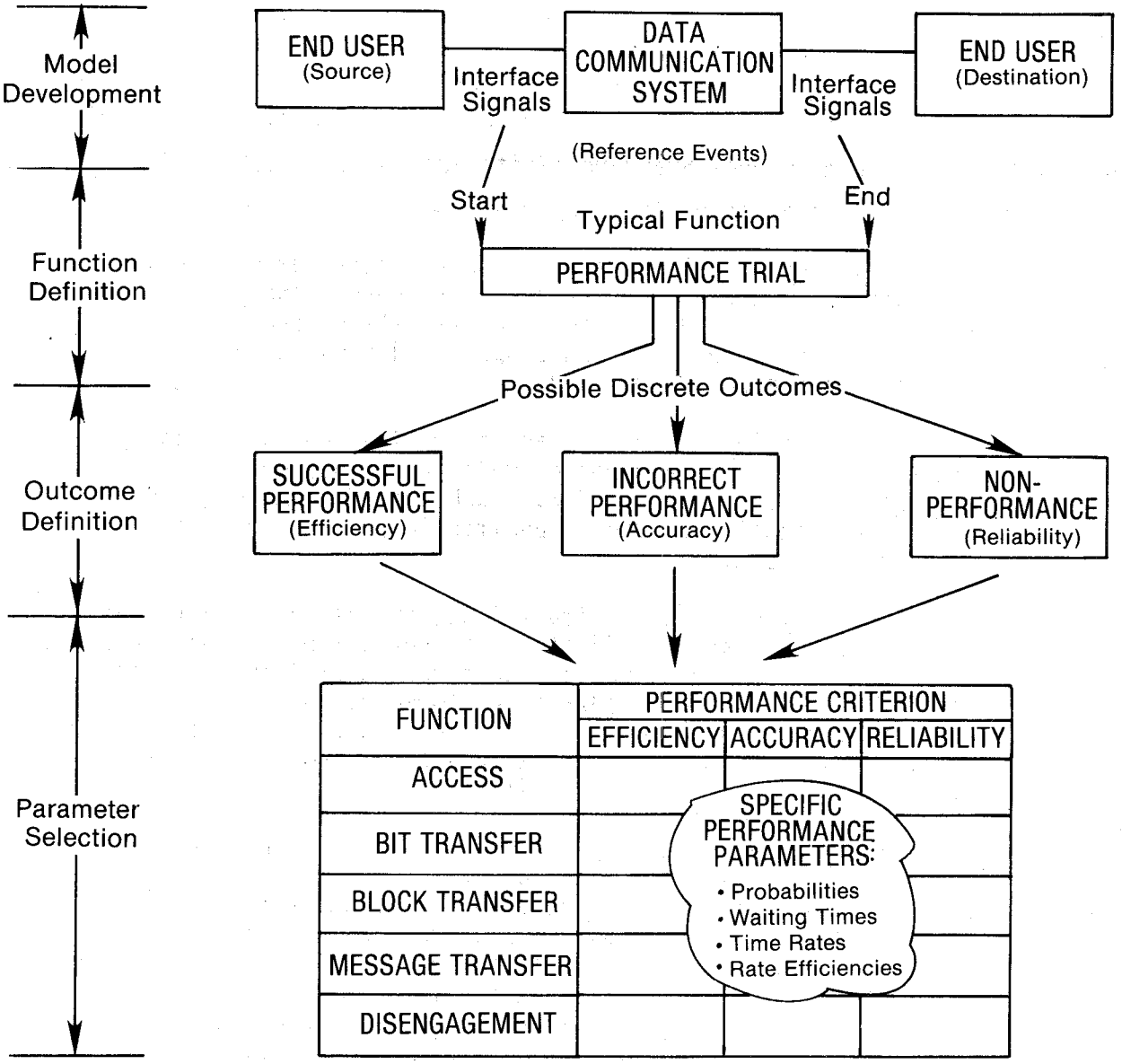


Figure 3-4. Parameter development overview.

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to the end user, and yet not restricted to a particular type of interface or a particular interaction sequence? How should the performance parameter definitions be related to such a description? These are questions which FED STD 1033 answers with the aid of a user-oriented telecommunication process model.

The model defines the end user of a telecommunication system or service as one or more of the following types of entities (Fig. 3-5):

1. A human terminal operator.
2. An unattended device medium (such as punched cards).
3. A computer application program performing data processing functions unrelated to telecommunication.

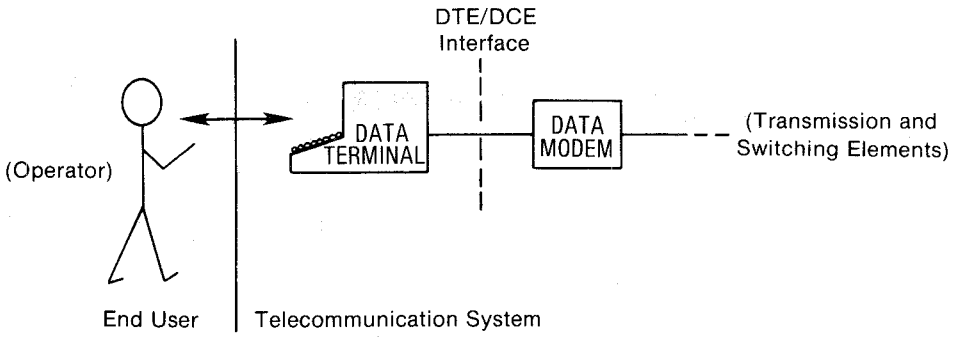
In some cases, more than one type of entity supports the overall user function: for example, a terminal operator providing control inputs and a punched paper tape providing information storage at the same data communication station.

In most traditional data communication configurations, the end user is an operator or medium, and there is a distinct physical unit, the data terminal, which converts information from user-readable form into "transmittable" form and vice versa. In all such cases, the telecommunication system is defined to include the data terminal and all elements of the information transfer channel on its line side. The user/system interface then corresponds to the physical interface between the operator or medium and the terminal (Figure 3-5a,b).

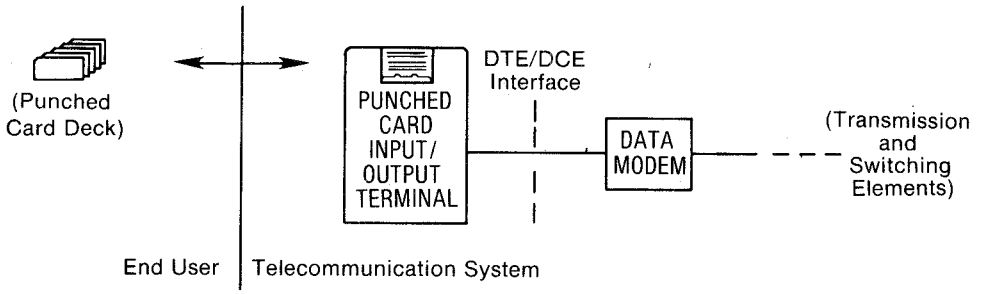
The development of computer communications and teleprocessing has given rise to configurations in which the end user is an application program within a digital computer. In most such cases, there is a separate program within the same computer (often called a "telecommunications access method") which functions as a first point of contact for application programs requiring telecommunication service. In all cases where such a program can be clearly identified, the telecommunication system is defined to include the access method and all functional and physical elements of the information transfer channel on its line side. The user/system interface then corresponds to the functional interface between the application program and the access method or its equivalent. One such interface, applicable to the so-called Open Systems Interconnection (OSI) model being developed by the International Standards Organization, is shown in Figure 3-5c (ISO, 1979).

The above definitions place the end user interfaces well outside the traditional DTE/DCE or "computer/communications" boundaries. This viewpoint is essential in a user-oriented standard, since modern terminals and high-level protocols

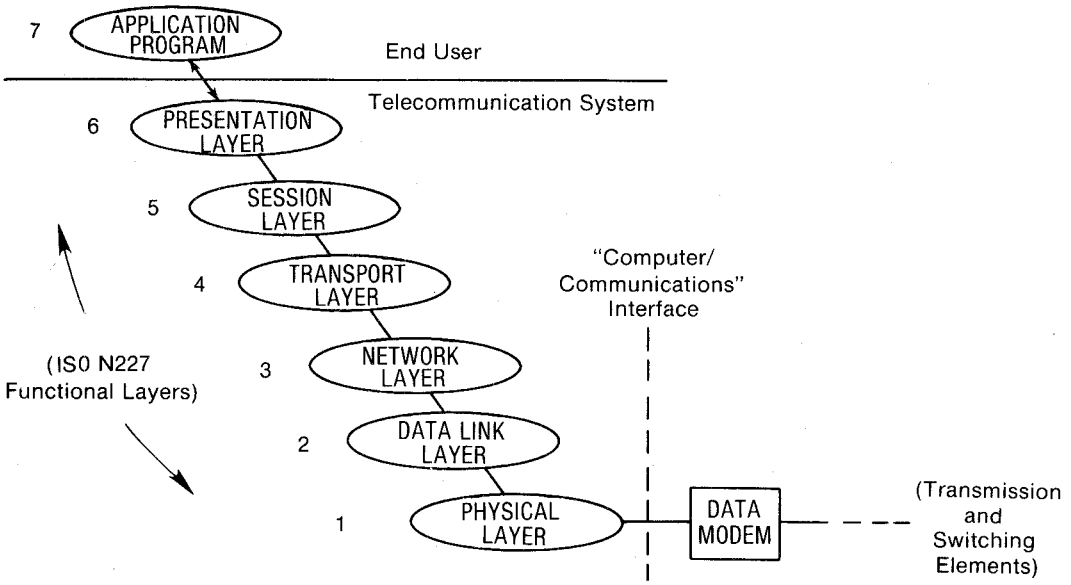
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a. Operator/Terminal Interface.



b. Medium/Terminal Interface.



c. Application Program/Access Method Interface.

Figure 3-5. User/system interfaces.

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perform communication functions (such as error control, flow control, and virtual circuit establishment) which have a profound effect on end-to-end performance. One modern data communication network whose end user interfaces are defined in this way is IBM's Systems Network Architecture (McFadyen, 1976).

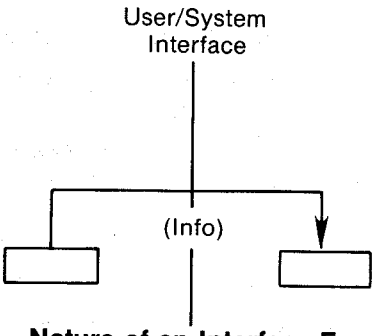
There are some computer communication and teleprocessing configurations in which it is difficult to draw a clear functional boundary between the "application program" and the "telecommunications access method." Guidelines for establishing appropriate user/system functional interfaces in such configurations are provided in Seitz and McManamon (1978).

Information can be transferred across user/system interfaces in a variety of ways. Typical interactions at the operator/terminals interface are manual keystrokes and the printing or displaying of received characters. Typical interactions at the medium/terminal interface are the reading and punching of punched cards. Typical interactions at the application program/access method interface are the issuance of system calls, co-routine calls, and inter-process communications, and the setting and clearing of flags. Information can be transferred between computer programs either by physical movement of computer words or by buffer reallocation (transfer of buffer "ownership" without physical movement of the stored information).

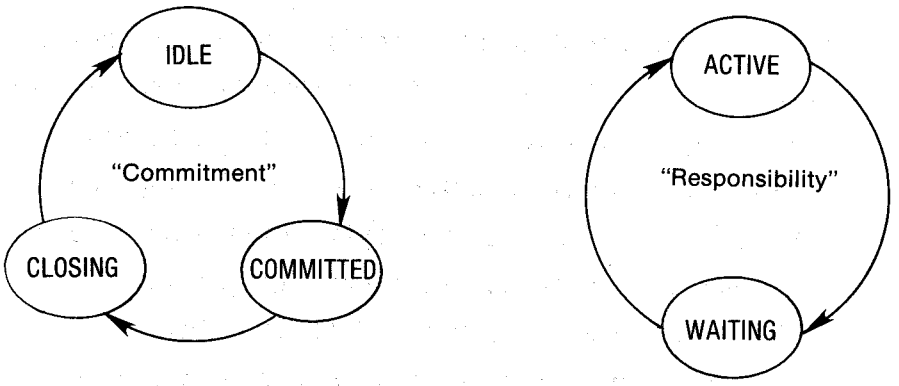
All of the user/system interactions just described are examples of what the standard calls "interface events" (Figure 3-6a). As defined in the standard, an interface event is a discrete transfer of user or overhead information from the physical possession and control of one entity (user or system) to that of the opposite entity (system or user). User information includes all information intended to cross both user/system interfaces. All other information (e.g., ANSI (1971), ENQ, ACK, and SYN characters, off-hook and on-hook signals) is overhead information.

In any description of performance, certain key interface events are identified as events to be counted, timed, or compared in calculating performance parameter values. As noted earlier, most existing standards and specifications identify such key events by reference to particular system-specific signals (e.g., off-hook). The FED STD 1033 model departs from this approach by defining the performance parameters in terms of more general, system-independent reference events. Each FED STD 1033 reference event is a "generic event" which subsumes many system-specific interface events having a common performance significance; and each is defined in such a way that it can always be identified, if it occurs, in any

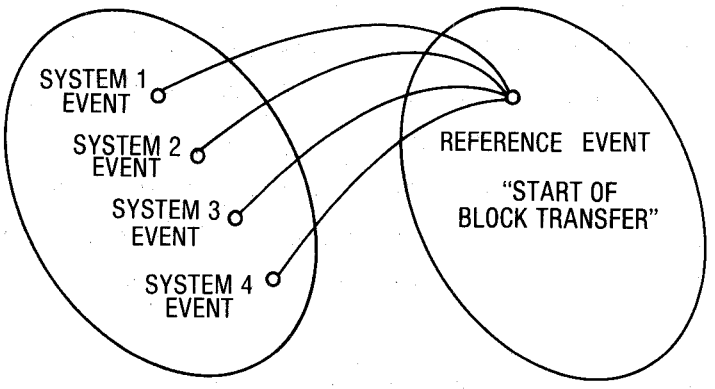
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a. Nature of an Interface Event.



b. Overhead Transfer Effects.



System	User Interface Device	System-Specific Event
1	Character-asynchronous TTY	Typing a letter or figure
2	Buffered CRT terminal	Typing CR (end of line)
3	Host computer access method	Issuing WRITE system call
4	Punched card reader	Reading of card column

c. Example - Reference Event "Start of Block Transfer".

Figure 3-6. Reference event concept.

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particular data communication transaction. The reference events collectively specify all information needed to describe performance in a comprehensive, user-oriented way.

The FED STD 1033 reference events are defined on the basis of the type of information transferred (overhead information or user information); the type of effect the transfer produces; and the particular user interface involved. Two basic types of overhead transfer effects are distinguished (Figure 3-6b):

1. Change in an entity's state of "commitment" to a particular information transfer transaction. Three possible commitment states are defined for each entity: Idle, Committed, and Closing.

2. Change in an entity's state of "responsibility" for creating the next transaction event at a particular user interface. Two possible responsibility states are defined for each entity: Active (responsible) and Waiting (not responsible).[9]

All overhead information transfers of significance to user-oriented performance assessment can be represented as changes in these basic transaction states; and no further information (other than event times) is needed to define the associated performance parameters. Thus, these transaction state changes are the reference events associated with overhead information transfers.

An example will help to clarify the relationship between system-specific interface events and the associated reference events. A user's action in lifting a telephone handset off-hook transfers one bit of overhead information (the new hookswitch position) from the user to the system. This system-specific interface event transfers the calling user (and the system) from the Idle state to the Committed state; and makes the system Active, and the user Waiting, relative to the next transaction event (dial tone). These commitment and responsibility state changes are the reference event that corresponds to going off-hook. If we know the nature and time of occurrence of this reference event, we need no further information about the system-specific event which generated it in order to define (for example) the start of the access function. The same reference event might be generated by a completely different interface event in another system: An example is issuance of a Connect system call in the ARPA network.

User information transfer events normally do not affect "commitment" or "responsibility" as described above; they simply move information from the

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[9]The model divides the system into two "half-system" entities to accommodate the possibility of independent effects at the two user interfaces.

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physical possession and control of one entity to that of another. Here again, certain key information must be specified to support performance assessment (e.g., user interface, information content, and event time); and that information can be specified in terms of system-independent reference events. Parameters describing transfer performance can then be defined on the basis of these system-independent events.

An example of a system-independent user information transfer event defined in FED STD 1033 is the Start of Block Transfer. Such an event must obviously be identified to define performance parameters such as Block Transfer Time and Block Transfer Rate. In order to define the reference event Start of Block Transfer, we must clearly identify (1) what is meant by a "block", and (2) when the transfer of a block between end users should be regarded as "started". FED STD 1033 defines a user information block as "a contiguous group of user information bits delimited at a source user/system interface for transfer to a destination user as a unit." The transfer of a block is said to have started when two conditions have been met:

1. The user information contained in the block is physically present within the system facility.

2. The system has been authorized to transmit that information.

The latter criterion (authorization) is the most natural way to establish the block boundaries as well; i.e., authorizing transmission of a given unit of information identifies that unit as a FED STD 1033 block. Authorization may either be an explicit user action (e.g., typing Carriage Return at a buffered CRT terminal) or an implicit part of inputting the user information itself (e.g., typing a single character at an asynchronous terminal).

Given the above definitions, the nature of the information unit called a "block" and the physical events associated with block transfer will differ widely from one system to another (Figure 3-6c). Nevertheless, in every system configuration some specific information unit can be identified as a FED STD 1033 block, and the start of transfer of that unit can be determined, using the above criteria. Hence, the reference event Start of Block Transfer can always be identified. A similar approach is used in defining the other user information transfer reference events.

In describing the need for the FED STD 1033 model, we raised two questions which have not been explicitly answered as yet:

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1. How can the process of telecommunication be described in a way that is meaningful and familiar to the end user, and yet not restricted to a particular type of interface or interaction sequence?

2. How should the performance parameter definitions be related to such a description?

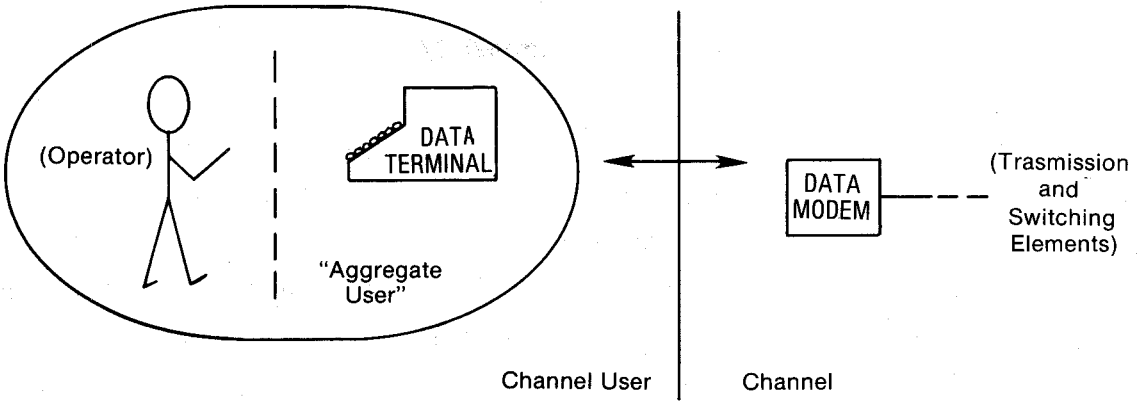
We are now in a position to answer these questions. The process of telecommunication should be described as a chronological sequence of reference events, each specifying the time and impact or content of an associated overhead or user information transfer. The performance parameter definitions should then be based on the information specified in such a system-independent event history. The FED STD 1033 model implements exactly such an approach.

Two concluding remarks regarding the FED STD 1033 model are appropriate at this point. The first deals with the question of relative detail. Readers of the standard will note that the model is presented there in somewhat more detail than was used here. A more complete presentation was needed in the standard to support the development of performance measurement procedures. Nevertheless, with a few minor additions provided later, the model information presented here is sufficient to understand the FED STD 1033 parameters and to apply them in performance specification.

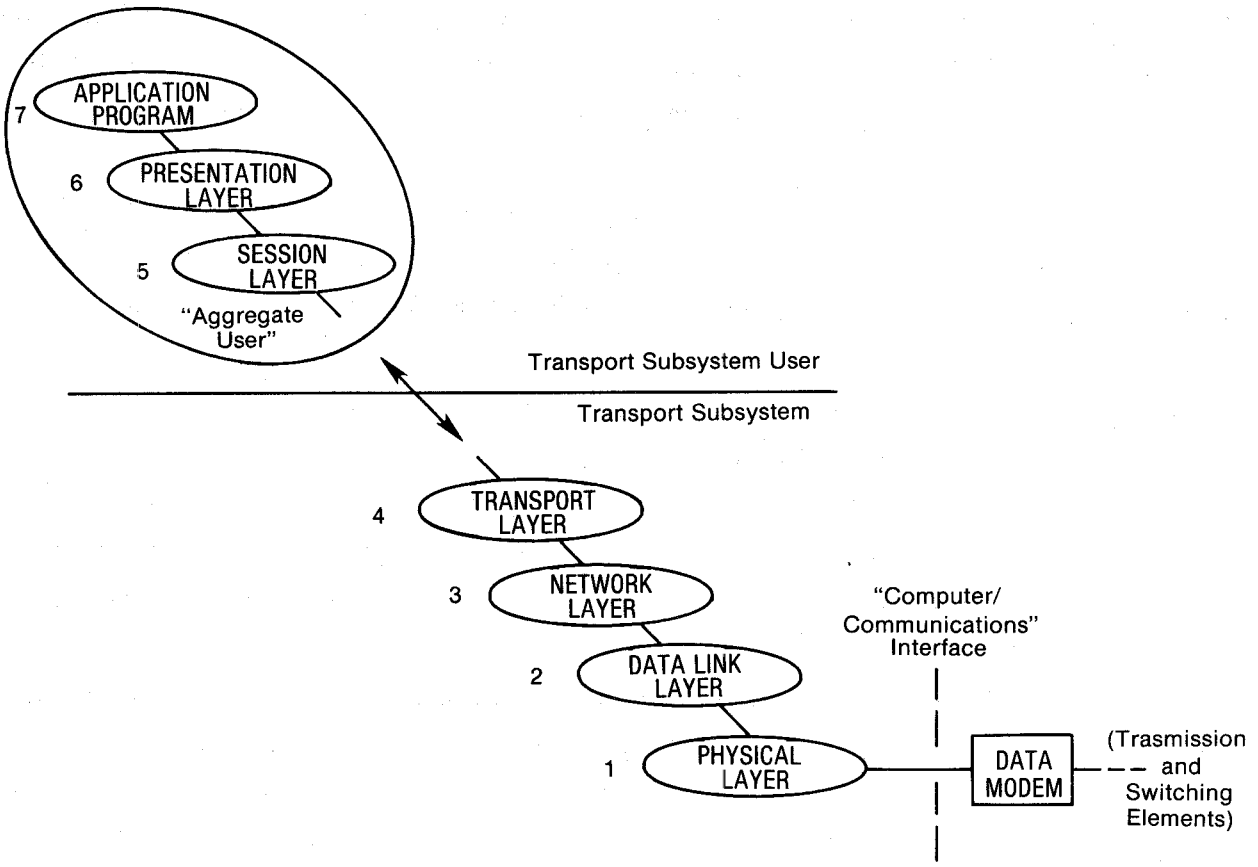
The second remark deals with alternative uses of the FED STD 1033 model. Although the model was developed primarily to represent end-to-end services, it is not restricted to such applications: any digital telecommunication process can be represented as a chronological sequence of reference events. In order to apply the model (and therefore the standard) to a digital subsystem, it is only necessary to (1) define the two interfaces of interest, (2) identify the specific events occurring at these interfaces, and (3) associate each specific interface event with a corresponding model reference event. The FED STD 1033 parameters can then be applied directly to the subsystem, since the parameter definitions are all based on the reference events.

Figure 3-7 illustrates two possible subsystem applications. In the first, Figure 3-7a, the subsystem interface is placed at the DTE/DCE physical interface; and the operator and terminal are regarded as an "aggregate user" of the information transfer channel. In the second, Figure 3-7b, the subsystem interface is placed between the Session and Transport layers of the OSI protocol hierarchy; and the Application, Presentation, and Session layers are regarded as an "aggregate user" of the Transport Subsystem (CCITT, 1980). Such applications can be useful

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a. Operator and Terminal as an "Aggregate User" of the Information Transfer Channel.



b. Application, Presentation and Session Layers as an "Aggregate User" of the Transport Subsystem.

Figure 3-7. Aggregate user concept.

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in allocating end-to-end performance objectives to purchasable components and services and, conversely, in determining the impact of subsystem choices on end-to-end performance.

3.3.2 Function Definition

Performance has little meaning as an isolated concept; to be useful, a description of performance must be clearly related to some particular function. The second step in developing the FED STD 1033 parameters was therefore to define a set of specific communication functions to be used as the focus of the performance description effort.

The five primary communication functions included in FED STD 1033 are defined in terms of particular model reference events as follows:

> The access function begins on issuance of an Access Request signal at the originating user interface, and ends (normally) on the next subsequent input of a source user information bit or block to the system. It encompasses all activities traditionally associated with physical circuit establishment (e.g., dialing, switching, ringing, modem handshaking) as well as any activities performed at higher protocol levels (e.g., X.25 virtual circuit establishment). Making the end of access coincident with the start of input of user information to the system reflects the user view that no data communication service has actually been provided until user information begins to flow.
>
> The bit, block, and message transfer functions describe the flow of user information between end users at three distinct levels of detail. Each function begins on the start of transfer of the associated information unit at the source user interface, and ends (normally) on completion of delivery of that unit to the intended destination. Each function encompasses all formatting, transmission, storage, error control, and media conversion activities performed between start of transfer and completion of delivery, including internal retransmissions if required. All three user information transfer functions must be considered in a comprehensive performance specification, as discussed below.
>
> The disengagement function begins on issuance of a Disengagement Request signal at either user interface, and ends (normally) on return of a corresponding Disengagement Confirmation signal. A separate disengagement function is defined for each end user; the two functions may either be linked (as in the case of circuit-switched systems) or independent (as in the case of message-switched systems).

The terms Access Request, Disengagement Request, and Disengagement Confirmation are general descriptors of purpose (i.e., names of reference events) rather than particular interface signals. An Access Request is any interface signal issued for the purpose of initiating an information transfer transaction. The

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corresponding commitment state change in the issuer is from Idle to Committed. Two specific Access Request signals have been cited earlier: The "off-hook" signal in the public switched network, and the Connect request in the ARPA network. Another typical Access Request signal is the Open Destination (OPNDST) VTAM Macro in IBM's Systems Network Architecture (SNA).

A Disengagement Request is any interface signal issued for the purpose of terminating an entity's participation in an information transfer transaction. The corresponding commitment state change in the issuer is from Committed to Closing. The Disengagement Request signals corresponding to the three Access Request signals just cited are the "on-hook" signal in the public switched network; the Close system call in the ARPA network; and the Close Destination (CLSDST) macro in SNA.

A Disengagement Confirmation is any interface signal issued for the purpose of confirming termination of an information transfer transaction. The corresponding commitment state change in the issuer from Closing to Idle. In the latter two systems cited above, Disengagement Confirmation is indicated by an explicit interface signal (completion flag). In the public switched network, Disengagement confirmation is an implied event which must be verified by subsequent user action (going "off-hook" and checking for dial tone).

The bit transfer, block transfer, and message transfer functions each serve a distinct purpose in the description of user information transfer performance. The bit transfer function fulfills the need for a "common denominator" to facilitate performance comparison - performance parameter values can always be compared at the bit level. The block transfer function describes performance relative to the information unit that is most relevant to the user in his internal operations - the user information block. The message transfer function provides a formal basis for defining the so-called "secondary" parameters, which describe the long-term availability of a data communication service. The "message" information unit is essentially a sample size, determined on the basis of measurement precision objectives as described in Crow and Miles (1976). ANSI Task Group X3S35 has adopted the term "sample" in preference to "message" to emphasize this fact.

The ANSI Task Group has also suggested a useful clarifying example in defining the end of the access function. In the case where the user interface device is a buffered CRT terminal, the end of access is defined to occur when the first user information character is typed, even though actual transmission of that character may not begin until the next subsequent carriage return is typed

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(authorizing transmission of the associated line block). The time difference between these two events can be quite substantial if the user's typing speed is slow, or if text editing is involved.

Figure 3-8 summarizes the intended interpretation of the standard with regard to the above points. The start of input of a block (Event 1) and authorization to transmit that block (Event 2) may or may not be coincident. If they are not, the end of access should be associated with the former event, and the start of block transfer with the latter. Note that the End of Block Transfer is associated with user notification (Event 4) in all cases.

An important characteristic of the primary communication functions defined in FED STD 1033 is that they are user dependent: i.e., their successful completion depends, in general, on events which must be produced by a user. As noted earlier, there is a problem in using parameters based on such functions to describe required system performance: the supplier has no control over user performance, and hence cannot ensure that user dependent parameter values will be met. FED STD 1033 overcomes this problem by explicitly describing the influence of user delay on the primary parameter values by means of separate "ancillary" parameters. The definition and use of these parameters is described in Section 3.3.5.

3.3.3 Outcome Definition

In defining performance parameters for a function, there is a clear need to identify the possible outcomes, or end results, that might occur on any given performance of that function. The third step in developing the FED STD 1033 parameters was to define such a set (or "sample space") of possible outcomes for each of the five primary communication functions. The possible outcomes for any given function can be grouped in three general categories:

1. Successful Performance. The function is completed within a specified maximum performance time, and the result or output is exactly what was intended. A familiar example is successful connection to the correct called party in a voice telephone call.

2. Incorrect Performance. The function is completed within the specified maximum performance time, but the result or output is somehow different from what was intended. A familiar example is connection to a "wrong number" (as a result of a system switching error) in a voice telephone call.

3. Nonperformance. The function is not completed within a specified maximum performance time. A familiar example is the blocking of a voice telephone call attempt by a circuit busy signal.

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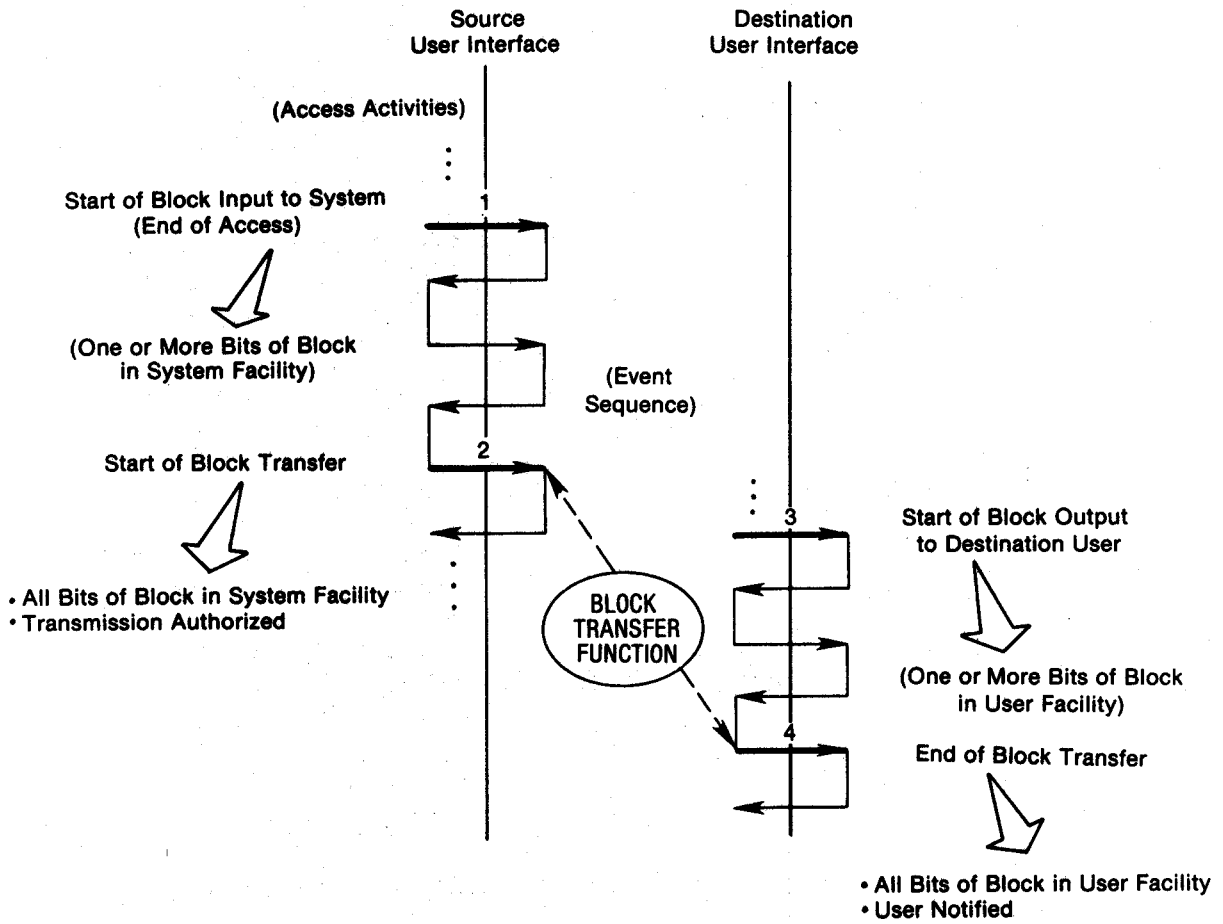


Figure 3-8. Block transfer function definition.

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These three outcome categories are significant because they correspond very closely with the three basic performance concerns (or "criteria") most frequently expressed by end users. Successful performance is associated with a user concern with efficiency or "speed"; i.e., in the case of successful performance, the user's concerns center on performance time or rate. Similarly, incorrect performance is associated with a user concern with accuracy; and nonperformance is associated with a user concern with reliability. FED STD 1033 uses these three general performance criteria as an overall framework for organizing the primary performance parameters.

Efficiency, accuracy, and reliability have extremely broad application in the assessment of performance. The questions "how fast, how accurately, and how reliably" apply to the performance of any function, irrespective of what the function "does" or how it is internally accomplished. Examples can be readily found in fields as diverse as energy conversion, manufacturing, transportation, and data processing, among others. The three criteria apply to user functions supported by communications as well as to communication functions, a fact that is helpful in developing "user requirements" for communication service. This subject will be discussed more fully in the Application Manual.

FED STD 1033 divides the Incorrect Performance and Nonperformance categories into more detailed outcomes to enable the definition of specific performance parameters. In general, the system outputs produced during the performance of a function can be "incorrect" in three ways: they may be incorrect in content, they may occur at an incorrect location, or they may include duplicate or other unwanted "extra" information. Failure to produce the expected output of a function can be a consequence, in general, of either system or user nonperformance. Thus, the standard distinguishes six possible outcomes of an individual "trial performance" of a typical primary function:

1. Successful Performance. The expected output occurs and is correct in both location (user interface) and content (delivered information).

2. Content Error. The expected output occurs at the correct location, but is incorrect in content.

3. Location Error. The expected output occurs at an incorrect location.

4. System Nonperformance. The expected output does not occur within the maximum performance time, as a result of either issuance of a blocking (busy) signal or excessive delay on the part of the telecommunication system.

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5. User Nonperformance. The expected output does not occur within the maximum performance time, as a result of either issuance of a blocking (busy) signal or excessive delay on the part of a user.

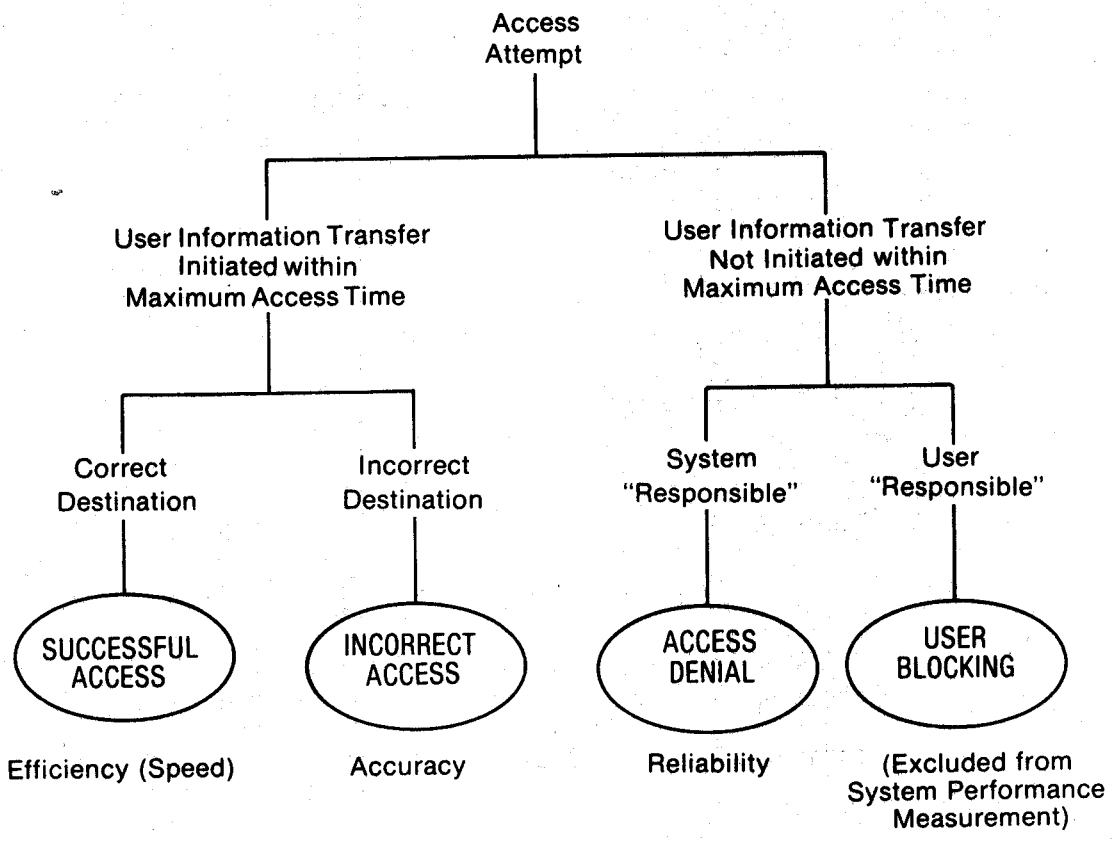
6. Extra Event. An unwanted (extra) output occurs in addition to that expected.

Outcome "sample spaces" for the five primary functions were defined by selecting pertinent outcomes from the above list, and specializing their meaning to the particular function in question. Figure 3-9 shows how this was done in the case of the access function. The standard defines four possible access outcomes: Successful Access, Incorrect Access, Access Denial, and User Blocking. Successful Access is the case where user information transfer is initiated as intended within a specified maximum access time. Incorrect Access is the case where transfer is initiated within the maximum time, but the transaction involves a user other than the one intended by the originator (i.e., a "wrong number"). Access Denial is the case where an access attempt fails as a result of either issuance of a blocking signal or excessive delay by the system. User Blocking is the case where an access attempt fails as a result of either issuance of a blocking signal or excessive delay by a user.

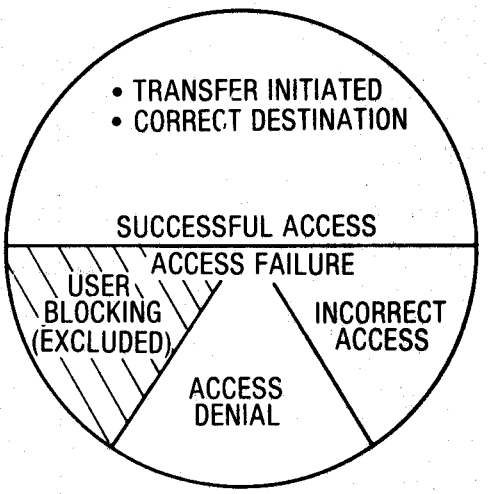
Familiar examples of system and user blocking signals are the "circuit busy" and "user busy" signals in the public switched network. User Blocking outcomes are excluded in defining the access performance parameters. Two of the six "typical function" outcomes defined earlier are not pertinent in the case of the access function: Content Error and Extra Event. In virtually all cases, such errors will result in either Incorrect Access or Access Denial.

Figure 3-10 shows the possible outcomes the standard defines for the block transfer function. Successful Block Transfer is the case where a transmitted block is delivered to the intended destination (within a specified maximum block transfer time), and the delivered block is completely correct in content. Incorrect Block is the case where a transmitted block is delivered to the intended destination, but the delivered block content includes one or more bit errors, additions, or deletions. Misdelivered Block is the case where a transmitted block is delivered to a destination other than that intended by the source. The block may be either correct or incorrect in content. Lost Block is the case where a transmitted block is not delivered to the intended destination within the maximum block transfer time, and the failure is attributable to the communication system. Refused Block is the case where a transmitted block is not delivered to the intended destination within the maximum block transfer time, and the failure is attributable to a

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a. Possible Outcomes of an Access Attempt.



b. Sample Space Representation.

Figure 3-9. Access outcome definition.

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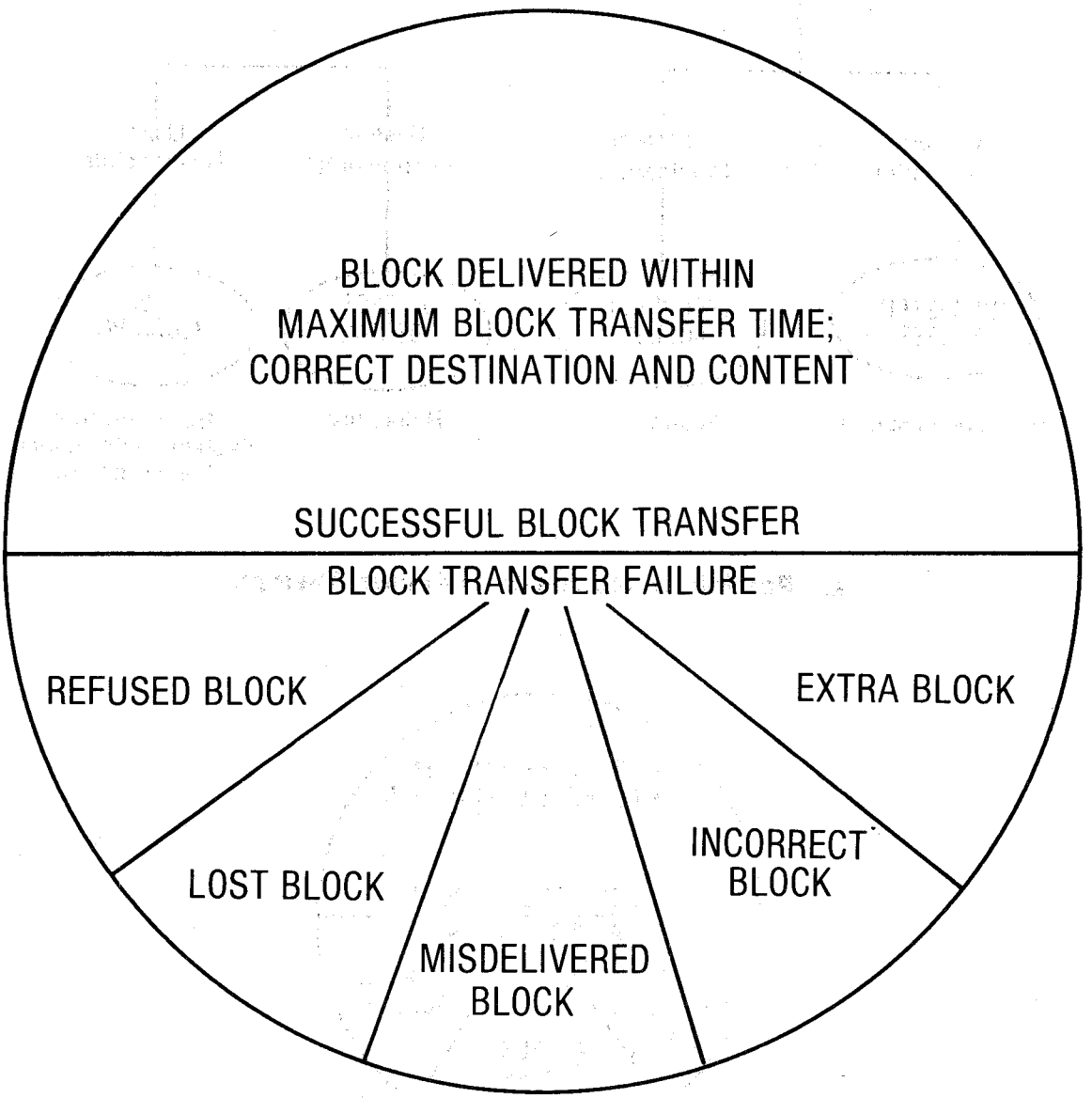


Figure 3-10. Block transfer outcomes.

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user.[10] Extra Block is the case where the system delivers to a destination a block that was not output by the source (e.g., a duplicate block).

FED STD 1033 defines the bit transfer and message transfer outcomes in a similar manner, and the corresponding "sample spaces" have the same form as that shown in Figure 3-10. The message transfer outcomes are basically collections of bit and block transfer outcomes; they are used in defining the "secondary" performance parameters as discussed in Section 3.3.5.

Figure 3-11 shows the possible outcomes the standard defines for the disengagement function. Successful Disengagement is defined to occur when the disengaging user and his local "half-system" are returned to the idle state (i.e., are freed to initiate a new transaction) within the specified maximum disengagement time. As noted earlier, this outcome is often indicated by an explicit Disengagement Confirmation signal issued by the system; but in some cases, it must be confirmed by a subsequent Access Request. Disengagement Denial is the case where the disengaging user (and his local half-system) are not returned to the idle state within the maximum disengagement time, and the failure is attributable to the communication system. User Disengagement Blocking is the case where the disengaging user (and his local half-system) are not returned to the idle state within the maximum disengagement time, and the failure is attributable to the user.

Figure 3-12 summarizes the possible outcomes (end results) FED STD 1033 defines for each of the five primary communication functions. Specific examples of each outcome are presented in Section 4.

3.3.4 Parameter Selection

The final step in developing the FED STD 1033 parameters was to select and define a minimum set of parameters to describe system performance relative to each function and outcome. Figure 3-13 illustrates how this was accomplished in the case of the access function. Access performance was described in terms of three specific parameters, one associated with each of the three general performance criteria noted earlier. The standard defines the selected access parameters essentially as follows.[11]

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[10] A destination user might "refuse" a block, for example, by failing to allocate necessary buffer space.

[11] Terminology and notation differ slightly from that used in the standard.

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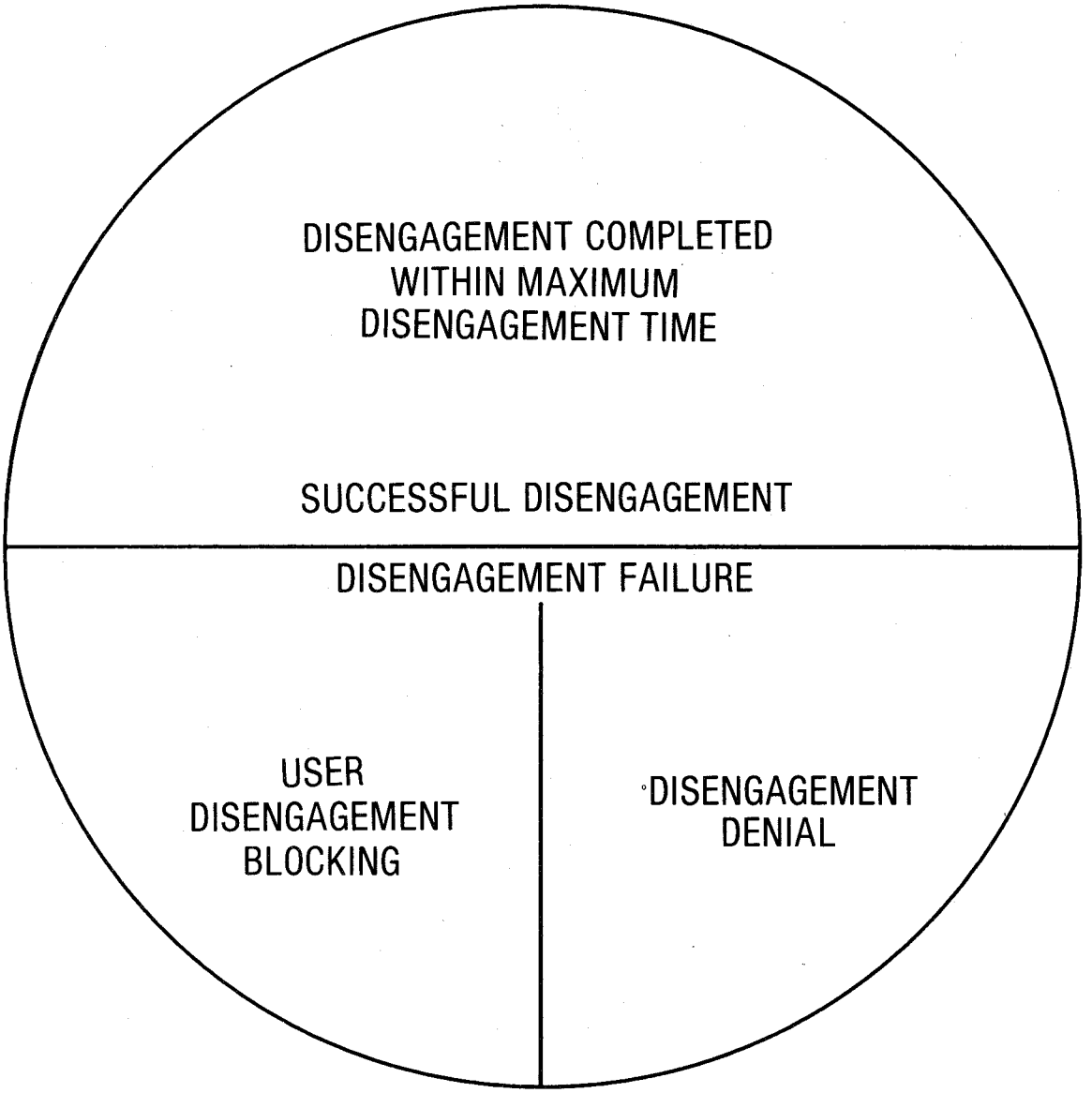


Figure 3-11. Disengagement outcomes.

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PRIMARY FUNCTIONS	OUTCOMES INCLUDED IN SAMPLE SPACE					
	SUCCESSFUL PERFORMANCE	CONTENT ERROR	LOCATION ERROR	SYSTEM NON-PERFORMANCE	USER NON-PERFORMANCE	EXTRA EVENT
ACCESS	√		√	√	√	
BIT TRANSFER	√	√	√	√	√	√
BLOCK TRANSFER	√	√	√	√	√	√
MESSAGE TRANSFER	√	√	√	√	√	√
DISENGAGEMENT	√			√	√	

Figure 3-12. Outcome summary.

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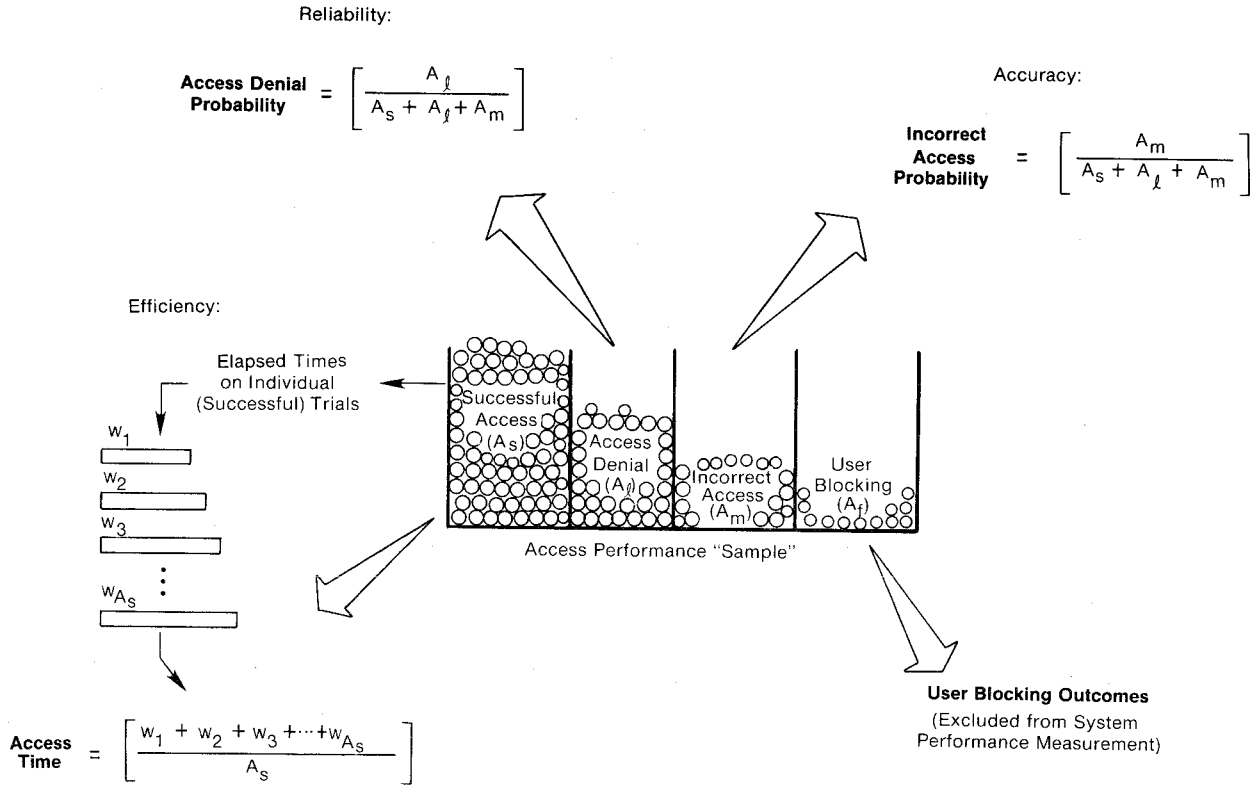


Figure 3-13. Access parameters.

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1. Access Time - Average value of elapsed time between the start of an access attempt and Successful Access. Elapsed time values are calculated only on access attempts that result in Successful Access.

2. Incorrect Access Probability - Ratio of total access attempts that result in Incorrect Access (i.e., connection to an unintended destination) to total access attempts included in an access performance sample (excluding User Blocking outcomes).

3. Access Denial Probability - Ratio of total access attempts that result in Access Denial (e.g., system blocking) to total access attempts included in an access performance sample (excluding User Blocking outcomes).

A key aspect of the FED STD 1033 parameter definitions is their expression in mathematical form. As noted earlier, this approach eliminates the ambiguity associated with purely narrative definitions, and also provides a standard procedure for calculating performance parameter values. The mathematical parameter definitions are based, in each case, on the concept of an access performance "sample" - i.e., a large number of successive access trials distributed, like apples, in appropriate outcome "bins". Each Successful Access outcome has an associated elapsed time value (the total time required to complete that particular attempt).

Values for the access parameters may be calculated directly from the data in an access performance sample. The value of the efficiency parameter Access Time is calculated by adding the individual elapsed times (w_i) for all A_s Successful Access outcomes, and then dividing by A_s. The value of the accuracy parameter Incorrect Access Probability is calculated by dividing the total number of Incorrect Access outcomes (A_m) by the total number of outcomes in the access sample, excluding the User Blocking outcomes - i.e., dividing A_m by $(A_s+A_\ell+A_m)$. Similarly, the value of the reliability parameter Access Denial Probability is calculated by dividing the number of Access Denial outcomes A_ℓ by $(A_s+A_\ell+A_m)$. User Blocking outcomes are excluded in calculating the access failure probabilities to ensure the comparability of values measured under different usage conditions.

The preceding section referred to a "maximum access time" beyond which an access attempt is declared a failure for performance assessment purposes. To ensure comparability, the standard fixes this "timeout" point at three times the Access Time specified for the service: i.e., three times the delay the user "expects to see" on any given access attempt. Note that this timeout constant has significance only in the assessment of performance; access attempts that extend beyond the timeout point need not be abandoned. Additional characteristics of the Access Time distribution (e.g., variance or 95-percent points) may also be of interest in some cases.

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The same general approach used in the access case was followed in selecting and defining performance parameters for the user information transfer and disengagement functions. A separate probability parameter was defined to express the likelihood of each possible failure outcome; and an "average elapsed time" parameter was defined, in each case, to quantify successful performance. Bit and Block Transfer Rate and Rate Efficiency parameters were also defined, to express performance from the standpoint of "throughput" and resource utilization. A complete list of the primary performance parameters specified in FED STD 1033 is provided in Section 3.3.6.

3.3.5 Secondary and Ancillary Parameters

Although the primary parameters described above provide a relatively detailed description of data communication performance, they fall short of completeness in two respects:

1. They do not provide the kind of macroscopic, long-term performance view users traditionally associate with the concept of availability.

2. They are user dependent, and thus cannot be applied directly in situations where it is necessary to describe unilateral system performance.

A small set of additional "secondary" and "ancillary" performance parameters were included in the standard to meet these needs.

Figure 3-14 illustrates the approach used in defining the secondary (availability) parameters. Very briefly, the sequence of transmissions between a specified pair of users is divided into a series of consecutive performance measurement periods or samples, each corresponding to the "message" information unit defined earlier. Values for each of five "supported" primary performance parameters are calculated on the basis of the outcome of each successive message transfer function. The calculated values are compared with corresponding outage thresholds to define the "secondary outcome" of that trial performance of the message transfer function as either Operational Service state or Outage state. Finally, appropriate time and probability parameters are defined to describe the resulting sequence of availability state transitions.

In assessing availability performance, the service connecting a user pair is observed only during the User Information Transfer (UIT) phase: i.e., the time, during each transaction, between Successful Access and disengagement of the last committed user (Seitz, 1980). There is no correspondence, in general, between the duration of an individual user information transfer phase and the length of a "message". As noted earlier, the "message" information unit is essentially a

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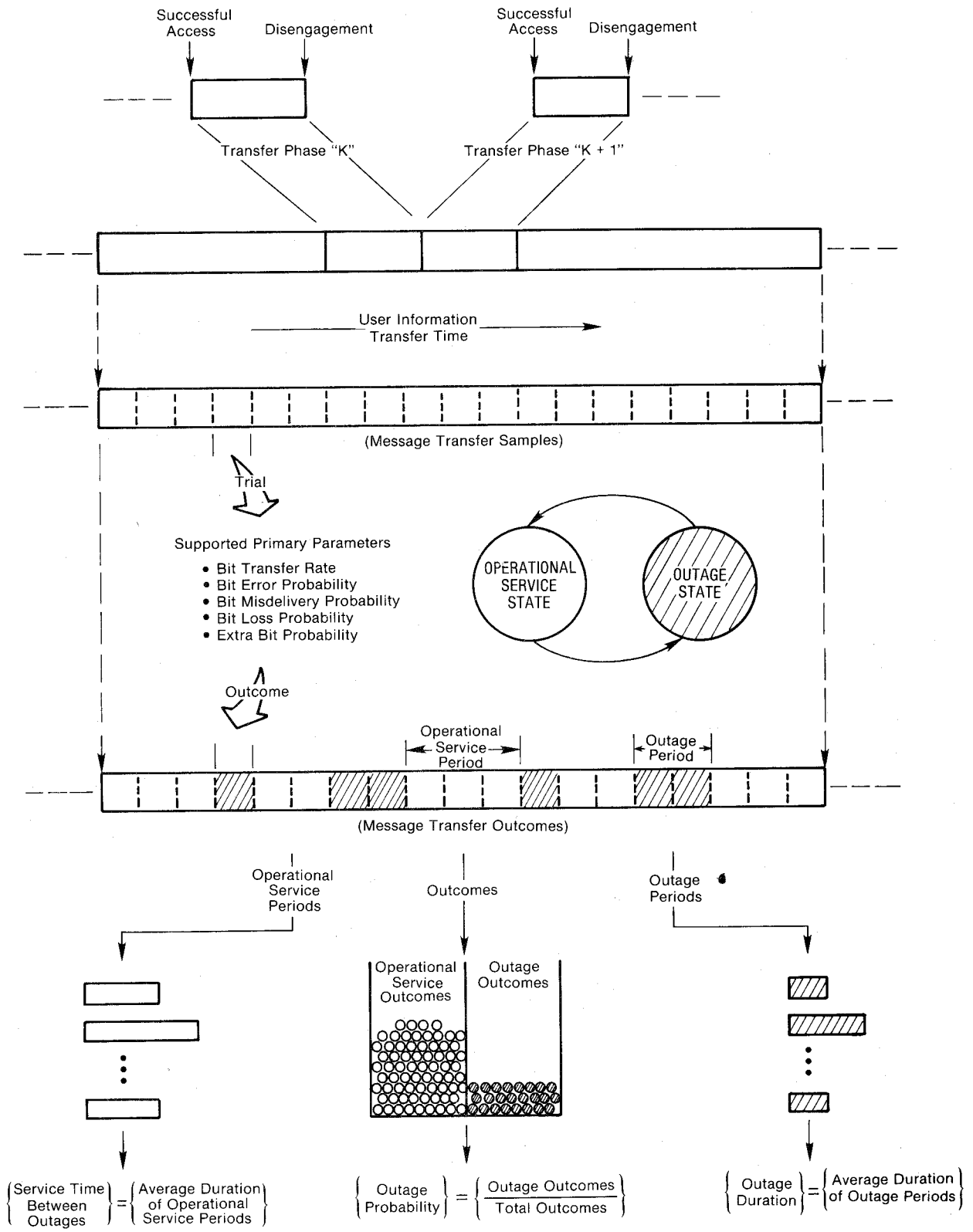


Figure 3-14. Secondary parameter development.

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sample size (a fixed number of transferred bits); and a given "message" may thus span more than one transaction.

Five primary user information transfer parameters are defined as supported performance parameters: the four bit transfer failure probabilities (Bit Error Probability, Bit Misdelivery Probability, Bit Loss Probability, and Extra Bit Probability) and Bit Transfer Rate. Outage thresholds for the supported performance parameters are defined as a function of the corresponding "nominal" values specified for the service as follows:

1. The outage threshold for Bit Transfer Rate is defined as one-third (1/3) of the nominal Bit Transfer Rate.

2. The outage thresholds for the four bit transfer failure probabilities are defined as a function of the corresponding nominal probability values by expressing the nominal value as a power of ten (for example, 10^{-6}) and then dividing the exponent by two (producing, for example, a threshold value of 10^{-3}). This procedure corresponds to taking the square root of the nominal (specified) probability value.

A service is defined to have been in the Operational Service state (during the preceding performance measurement period) whenever the measured values for all supported parameters are better than their associated outage thresholds. A service is defined to have been in the Outage state whenever the measured values for one or more supported parameters are worse than their associated thresholds. This definition process produces (in the measurement record) a sequence of alternating Operational Service and Outage periods, each having a known duration in the User Information Transfer (UIT) time. Each period comprises an integer number of messages or samples. The secondary performance parameters provide a statistical description of this two-state random process. They are defined in the standard essentially as follows.

Service Time Between Outages - Average value of elapsed User Information Transfer time between entering and next leaving the Operational Service state.

Outage Duration - Average value of elapsed User Information Transfer time between entering and next leaving the Outage state.

Outage Probability - Ratio of total message transfer attempts resulting in the Outage state to total message transfer attempts included in a secondary parameter measurement.

These parameters are termed "secondary" to emphasize the fact that they are defined on the basis of measured primary parameter values, rather than on the basis of direct observations of interface events.

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The final category of FED STD 1033 parameters to be described are the ancillary parameters. As discussed earlier, the primary communication functions defined in FED STD 1033 are, in general, user dependent; and there is a need, then, for a quantitative means of expressing the influence of user delay on the primary parameter values. The ancillary parameters fulfill that need.

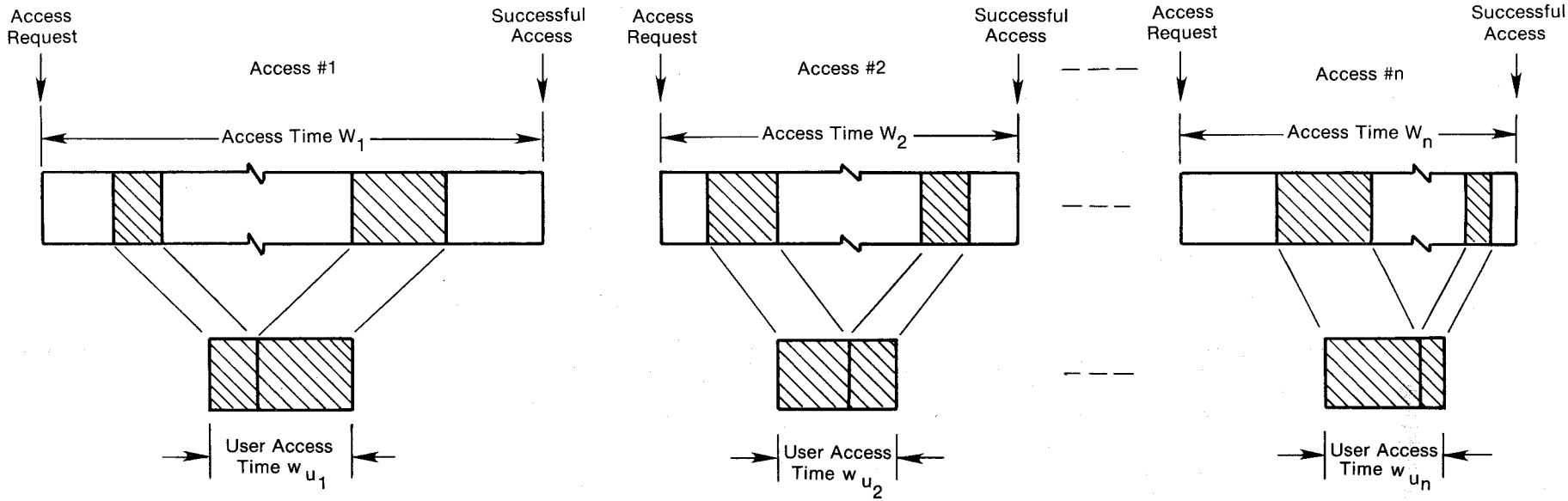
Very briefly, the ancillary parameters are developed by dividing the total performance time for an associated primary function into alternating periods of system and user "responsibility"; and then calculating the average proportion of total performance time for which the users are "responsible". As a simple illustration, consider the voice telephone access example discussed earlier (Fig. 3-3). The total performance time for the access function is the time between the calling user's off-hook action and the called party's answer. This total performance time can be divided into alternating periods of system and user responsibility by noting, at any time, which entity must produce the next interface event. During the period between off-hook and dial tone, the system is responsible; during the period between dial tone and positioning of the first dialed digit, the user is responsible; and so on. The ancillary parameter User Access Time Fraction expresses the average proportion of total Access Time attributable to the user activities.

Figure 3-15 illustrates the approach used in defining the ancillary parameters in more detail, again using the primary function of access as an example. The figure depicts a series of successful access attempts, each having a total access time w and a total user access time w_u. The latter quantity represents the total access time attributable to user responsibility on each particular trial. The ancillary parameter User Access Time Fraction is calculated by adding the user access time values over a suitable number of successful access attempts, and then dividing by the corresponding sum of the total access times. Only Successful Access outcomes are considered in estimating User Access Time Fraction in order to avoid biasing the average with unrepresentative values.

A similar approach is used in defining ancillary performance parameters for the block transfer, message transfer, and disengagement functions. No ancillary parameter is defined for the bit transfer function, since its values can be inferred from the corresponding block transfer parameter. The standard thus defines a total of four ancillary performance parameters: User Access Time Fraction, User Block Transfer Time Fraction, User Message Transfer Time Fraction, and User Disengagement Time Fraction.

In describing the ancillary parameters, it should be noted that there are cases where it is not possible to place unilateral responsibility for completing a

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$$\left\{\begin{array}{c}\text{User Access}\\ \text{Time Fraction}\end{array}\right\} = \left\{\frac{\text{Average User Access Time}}{\text{Average Total Access Time}}\right\}$$

$$= \left\{\frac{w_{u_1} + w_{u_2} + \ldots + w_{u_n}}{W_1 + W_2 + \ldots + W_n}\right\}$$

Figure 3-15. Ancillary parameter development - access example.

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function on the user or the system. Activities may proceed concurrently at the two interfaces, with the user responsible at one and the system responsible at the other. The message transfer function provides a straightforward example. A "message" typically consists of many separate blocks. At a given moment during transfer of a message, the source user may be inputting one block to the system while the system is outputting a previous block to the destination. Neither entity is solely responsible, at that moment, for the overall function of message transfer; responsibility is "split" equally between the two.

FED STD 1033 accommodates such concurrency by defining responsibility separately for each user interface, and counting intervals of split responsibility at half their actual value in calculating total user performance time. Thus, if a message transfer interval included 2 minutes of unilateral user responsibility and 2 minutes of split responsibility, the total message transfer time attributed to the users would be 3 minutes. (Split responsibility is nonexistent in many data communication transactions, and it can often be disregarded with negligible effect on a performance specification.)

The ancillary parameters have two specific uses:

1. They enable calculation of "user-independent" values for the associated efficiency parameters - i.e., the values that would observed if all user delays were zero.

2. They provide a basis for identifying the entity "responsible" for timeout failures - the user or the system.

Each of these uses is described more fully in Section 4.6.

3.3.6 Problem Solutions - Summary

We noted earlier that the development of FED STD 1033 required the solution of three key technical problems. The technical approach adopted in the standard provides a solution to each of these problems, as summarized below.

1. System Dependence. The standard solves this problem through the expedient of the user-oriented performance model. The model reduces all user/system interactions to a small set of general reference events which can be identified in any system; and the performance parameter definitions are then based on these system-independent events.

2. Detailed Parameter Definition. The standard solves this problem by using sample spaces and mathematical equations as the major parameter definition tools. Sample spaces encourage the analyst to consider, and carefully define, all relevant outcomes of a performance trial. Equation definitions eliminate the ambiguity often associated with purely narrative definitions.

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3. User Dependence. The standard solves this problem through the use of the ancillary performance parameters. These parameters provide a basis for "factoring out" user influence on the waiting time, time rate, and rate efficiency parameters; and a means of determining whether the user or the system is "responsible" for timeout failures.

Figure 3-16 summarizes the performance parameters ultimately selected for inclusion in Interim Federal Standard 1033. A total of 26 parameters were selected, including 19 primary parameters, 3 secondary parameters, and 4 ancillary parameters. Each selected parameter is described in detail in the following section.

4. UNDERSTANDING THE PARAMETERS

4.1 Introduction

Suppose you were handed the parameter table of Figure 3-16, with no prior explanation, and asked to use it in specifying a communication service requirement. What questions would you ask about each parameter before beginning the specification? For most potential users, the key questions about each parameter would include the following:

- What is the meaning of this parameter, in simple, straightforward, user-oriented terms? How is it related to other widely-used performance parameters?
- Why is the value of this parameter significant to data communications users? What are its best and worst possible values, and what are their implications?
- What typical values might be specified for this parameter, in characterizing (a) performance requirements for familiar user applications, and (b) performance capabilities of existing data communication systems and services?
- How do the values for this parameter influence, and how are they influenced by, the key decisions in communication system design?

This section answers these questions by means of tutorial "essay descriptions" of the FED STD 1033 parameters. The individual parameter essays are organized by function and category in the manner suggested above: i.e., access parameters, user information transfer parameters, disengagement parameters, secondary parameters, and ancillary parameters. A separate essay is provided for each primary parameter, with the exception that corresponding bit- and block-oriented transfer parameters (e.g., Bit Error Probability - Block Error Probability) are described together. The secondary parameters and the ancillary parameters are each described

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